Ketorolac (systemic): Drug information - UpToDate

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Contributor Disclosures

For additional information see ”Ketorolac (systemic): Patient drug information” and ”Ketorolac (systemic): Pediatric drug information”

For abbreviations, symbols, and age group definitions show table

ALERT: US Boxed Warning

Appropriate use:

Ketorolac is indicated for the short-term (up to 5 days in adults) management of moderately severe acute pain that requires analgesia at the opioid level. Oral ketorolac is only indicated as continuation treatment following IV or IM dosing of ketorolac, if necessary. The total combined duration of use of ketorolac tablets and injection should not exceed 5 days. The recommended total daily dose of ketorolac tablets (maximum 40 mg) is significantly lower than for ketorolac injection (maximum 120 mg).

Ketorolac is not indicated for use in pediatric patients and is not indicated for minor or chronic painful conditions. Increasing the dose of ketorolac beyond labeled recommendations will not provide better efficacy but will increase the risk of developing serious adverse events.

Cardiovascular thrombotic events:

Nonsteroidal anti-inflammatory drugs (NSAIDs) cause an increased risk of serious cardiovascular thrombotic events, including myocardial infarction and stroke, which can be fatal. This risk may occur early in treatment and may increase with duration of use.

Ketorolac is contraindicated during the perioperative setting of coronary artery bypass graft surgery.

GI risk:

Ketorolac can cause peptic ulcers, GI bleeding, and/or perforation of the stomach or intestines, which can be fatal. These events can occur at any time during use and without warning symptoms. Therefore, ketorolac is contraindicated in patients with active peptic ulcer disease, recent GI bleeding or perforation, and a history of peptic ulcer disease or GI bleeding. Elderly patients and patients with a prior history of peptic ulcer disease and/or GI bleeding are at greater risk for serious GI events.

Intrathecal or epidural administration:

Ketorolac is contraindicated for intrathecal or epidural administration due to its alcohol content.

Hypersensitivity:

Hypersensitivity reactions, ranging from bronchospasm to anaphylactic shock, have occurred and appropriate counteractive measures must be available when administering the first dose of ketorolac injection. Ketorolac is contraindicated in patients with previously demonstrated hypersensitivity to ketorolac or allergic manifestations to aspirin or other NSAIDs.

Renal risk:

Ketorolac is contraindicated in patients with advanced renal impairment and in patients at risk for renal failure due to volume depletion.

Risk of bleeding:

Ketorolac inhibits platelet function and is, therefore, contraindicated in patients with suspected or confirmed cerebrovascular bleeding, hemorrhagic diathesis, incomplete hemostasis, and those at high risk of bleeding.

Ketorolac is contraindicated as a prophylactic analgesic before any major surgery.

Risk during labor and delivery:

The use of ketorolac in labor and delivery is contraindicated because it may adversely affect fetal circulation and inhibit uterine contractions.

Concomitant use with nonsteroidal anti-inflammatory drugs:

Ketorolac is contraindicated in patients currently receiving aspirin or NSAIDs because of the cumulative risks of inducing serious NSAID-related adverse events.

Special populations:

Dosage should be adjusted for patients ≥65 years of age, for patients <50 kg (110 lbs) of body weight, and for patients with moderately elevated serum creatinine. Doses of ketorolac injection are not to exceed 60 mg (total dose per day) in these patients.

Brand Names: Canada

- ALTI-Ketorolac;
- APO-Ketorolac;
- JAMP Ketorolac;
- Mar-Ketorolac;
- MINT-Ketorolac;
- Toradol

Pharmacologic Category

- Analgesic, Nonopioid;
- Nonsteroidal Anti-inflammatory Drug (NSAID), Oral;
- Nonsteroidal Anti-inflammatory Drug (NSAID), Parenteral

Dosing: Adult

Dosage guidance:

Safety: Consider proton pump inhibitor coadministration in patients at risk for GI bleeding (eg, taking dual antiplatelet therapy or an anticoagulant, ≥60 years of age) (Ref). May increase cardiovascular risk (eg, thrombotic events), particularly in patients with established cardiovascular disease.

Dosing: Use the lowest effective dose for the shortest duration of time; do not shorten dosing interval of 4 to 6 hours. Patients weighing <50 kg or ≥65 years of age may be at increased risk for adverse effects; lower doses are recommended.

Migraine, severe, acute treatment

Migraine, severe, acute treatment (off-label use):

Weight ≥50 kg and <65 years of age:

IV: 30 mg once (Ref).

IM: 60 mg once (Ref).

Weight <50 kg or ≥65 years of age:

IV: 15 mg once (Ref).

IM: 30 mg once (Ref).

Pain management

Pain management, acute:

Note: Do not exceed recommended doses or administer with another nonsteroidal anti-inflammatory drug (NSAID). May supplement with analgesic(s) from another therapeutic class for breakthrough pain. Oral formulation should not be given as initial therapy; other better tolerated oral NSAIDs are generally preferred.

Weight ≥50 kg and <65 years of age:

IV: 30 mg as a single dose or 15 to 30 mg every 6 hours as needed; maximum daily dose: 120 mg/day; maximum duration: 5 days combined (parenteral, oral, and nasal) (Ref).

IM: 30 to 60 mg as a single dose or 15 to 30 mg every 6 hours as needed; alternatively, may administer 10 to 30 mg every 4 to 6 hours as needed; maximum daily dose: 120 mg/day; maximum duration: 5 days combined (parenteral, oral, and nasal) (Ref).

Oral (only as continuation of IM or IV therapy, not for initial therapy): 20 mg once, followed by 10 mg every 4 to 6 hours as needed; maximum daily dose: 40 mg/day; maximum duration: 5 days combined (parenteral, oral, and nasal) (Ref).

Weight <50 kg or ≥65 years of age:

IV: 15 mg as a single dose or 15 mg every 6 hours as needed; maximum daily dose: 60 mg/day; maximum duration: 5 days combined (parenteral, oral, and nasal) (Ref).

IM: 30 mg as a single dose or 15 mg every 6 hours as needed; alternatively, may administer 10 mg every 4 to 6 hours as needed; maximum daily dose: 60 mg/day; maximum duration: 5 days combined (parenteral, oral, and nasal) (Ref).

Oral (only as continuation of IM or IV therapy, not for initial therapy): 10 mg every 4 to 6 hours as needed; maximum daily dose: 40 mg/day; maximum duration: 5 days combined (parenteral, oral, and nasal) (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

The renal dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

Altered kidney function:

eGFR ≥60 mL/minute/1.73 m^2: **IM, IV, Oral:** No dosage adjustment necessary (Ref).

eGFR >30 to <60 mL/minute/1.73 m^2:

IM, IV: Use of analgesics other than nonsteroidal anti-inflammatory drugs may be preferred. If necessary, consider 7.5 to 15 mg every 6 hours (Ref); use the lowest effective dose for the shortest duration possible. Avoid use in patients at high risk for acute kidney injury (ie, volume depleted, hypotensive, elderly, or taking concurrent nephrotoxic medications) (Ref).

Oral: 10 mg every 4 to 6 hours as needed; maximum: 40 mg/day; oral dosing is intended to be a continuation of IM or IV therapy (Ref).

eGFR ≤30 mL/minute/1.73 m^2: **IM, IV, Oral:** Avoid use due to increased risk of acute kidney injury (Ref). The manufacturer’s labeling states use is contraindicated in advanced kidney impairment.

Hemodialysis, intermittent (thrice weekly): Not dialyzable: **IM, IV, Oral:** Avoid use, as patients with end-stage kidney disease may be at increased risk for bleeding (eg, GI), cardiovascular adverse effects, and loss of residual kidney function (Ref). The manufacturer's labeling states use is contraindicated in advanced kidney impairment.

Peritoneal dialysis: Unlikely to be significantly dialyzed given very high protein binding (Ref): **IM, IV, Oral:** Avoid use, as patients with end-stage kidney disease may be at increased risk for bleeding (eg, GI), cardiovascular adverse effects, and loss of residual kidney function (Ref). The manufacturer's labeling states use is contraindicated in advanced kidney impairment.

CRRT: Avoid use (Ref).

PIRRT (eg, sustained, low-efficiency diafiltration): Avoid use (Ref).

Dosing: Liver Impairment: Adult

The liver dosing recommendations are based upon the best available evidence and clinical expertise. Senior Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

Note: Nonsteroidal anti-inflammatory agents inhibit the prostaglandins required in cirrhosis to mitigate the effects of the renin-angiotensin-aldosterone system on the renal vasculature and cyclooxygenases in the GI tract. Ketorolac can precipitate kidney insufficiency and/or increase the risk of upper GI bleeding, especially in the setting of cirrhosis (Ref).

Liver impairment prior to treatment initiation:

Initial or dose titration in patients with preexisting liver cirrhosis:

Child-Turcotte-Pugh class A to C: Avoid use (Ref).

Liver impairment developing in patient *already receiving ketorolac:*

Ketorolac-induced liver injury: Permanently discontinue use if ketorolac-induced liver injury is confirmed and laboratory abnormalities cannot be explained by other etiologies (eg, alcohol, active viral hepatitis) (Ref).

Dosing: Older Adult

Avoid use (Ref).

Dosing: Pediatric

(For additional information see "Ketorolac (systemic): Pediatric drug information")

Dosage guidance:

Dosing: To reduce the risk of adverse cardiovascular and GI effects, use the lowest effective dose for the shortest period of time. The total duration of ketorolac (combined IV/IM and oral) should not exceed 5 days (Ref).

Pain management

Pain management (acute; moderately severe):

Infants and Children <2 years: Limited data available: IV: 0.5 mg/kg/dose every 6 to 8 hours, not to exceed 48 to 72 hours of treatment; has been used postoperatively primarily following cardiac and abdominal surgery; maximum dose: 15 mg/dose (Ref). A lower dose of 0.25 mg/kg/dose every 6 to 8 hours has also been recommended (Ref). **Note:** IM administration has been described in patients ≥6 months of age (Ref).

Children ≥2 years and Adolescents ≤16 years: Limited data available:

IM, IV: 0.5 mg/kg/dose every 6 to 8 hours; maximum dose: 30 mg/dose, usual reported duration: 48 to 72 hours; not to exceed 5 days of treatment (Ref).

Oral: 1 mg/kg/dose every 4 to 6 hours; maximum dose: 10 mg/dose; maximum daily dose: 40 mg/**day** (Ref). **Note:** Oral formulation should only be used as continuation of IV or IM therapy; do not use as initial therapy (Ref).

Adolescents ≥17 years: **Note:** The maximum combined duration of treatment (for parenteral and oral) is 5 days; do not increase dose or frequency; supplement with low-dose opioids if needed for breakthrough pain.

<50 kg:

IM: 30 mg as a single dose or 15 mg every 6 hours; maximum daily dose: 60 mg/**day**.

IV: 15 mg as a single dose or 15 mg every 6 hours; maximum daily dose: 60 mg/**day**.

Oral: Initial: 10 mg, then 10 mg every 4 to 6 hours; maximum daily dose: 40 mg/**day**.

≥50 kg:

IM: 60 mg as a single dose or 30 mg every 6 hours; maximum daily dose: 120 mg/**day**.

IV: 30 mg as a single dose or 30 mg every 6 hours; maximum daily dose: 120 mg/**day**.

Oral: Initial: 20 mg, then 10 mg every 4 to 6 hours; maximum daily dose: 40 mg/**day**.

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no pediatric-specific dosage adjustments provided in the manufacturer's labeling; however, dosage adjustments are recommended for adult patients. The specific degree of renal impairment where use is permitted is not defined in the product labeling; however, use is contraindicated in patients with advanced renal impairment or those at risk for renal failure due to volume depletion; some experts have suggested the following:

Infants, Children, and Adolescents:

Aronoff 2007 recommendations: **Note:** Renally adjusted dose recommendations are based on doses of IV, IM: 0.25 to 1 mg/kg/dose every 6 hours (Ref).

GFR ≥30 mL/minute/1.73 m^2: No dosage adjustment necessary.

GFR 10 to 29 mL/minute/1.73 m^2: Administer 50% of dose.

GFR <10 mL/minute/1.73 m^2: Administer 25% to 50% of dose.

Intermittent hemodialysis: Avoid use.

Peritoneal dialysis: Avoid use.

KDIGO guidelines provide the following recommendations for NSAIDs (Ref):

eGFR 30 to <60 mL/minute/1.73 m^2: Temporarily discontinue in patients with intercurrent disease that increases risk of acute kidney injury.

eGFR <30 mL/minute/1.73 m^2: Avoid use.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling. Use with caution, may cause elevation of liver enzymes; discontinue if clinical signs and symptoms of liver disease develop.

Adverse Reactions (Significant): Considerations

Cardiovascular effects

Use of nonsteroidal anti-inflammatory drugs (NSAIDs) is associated with an increased risk of serious adverse cardiovascular (CV) events, including **acute myocardial infarction** (MI), **cerebrovascular accident**, and CV death. New-onset **hypertension** or exacerbation of hypertension may occur with NSAID use, which may also contribute to an increased risk of CV events; this effect was documented following use of celecoxib, ibuprofen, or naproxen in the PRECISION-ABPM trial (Ref). NSAIDS should be avoided in patients hospitalized for acute coronary syndromes due to the risk of adverse cardiac events (Ref). New-onset or exacerbation of **heart failure** may also occur with cyclooxygenase (COX)-2 selective NSAIDs (ie, coxibs) and nonselective NSAIDs, including ketorolac, resulting in an increased risk of hospitalizations for heart failure and death in patients with heart failure (Ref).

Data collected by the Coxib and traditional NSAID Trialists' Collaborative have shown that high-dose naproxen may have the most favorable CV risk profile among NSAIDs analyzed (Ref); however, data from the PRECISION trial showed no difference with regards to risk between naproxen, ibuprofen, or celecoxib after a treatment duration of therapy of ~3 years (Ref).

CV risk associated with use of ketorolac was not evaluated in long-term studies as treatment with ketorolac is limited to short-term use (ie, 5 days). Additionally, several retrospective studies evaluating the use of ketorolac in patients with cardiovascular disease (CVD) undergoing coronary artery bypass graft (CABG) surgery have failed to find an increased risk of adverse effects including mortality, MI, and bleeding events in this patient population (Ref); clinicians should note that these studies cannot confirm mortality benefit and/or increased graft patency. The FDA states that there are insufficient data to determine if the risk of MI or stroke is definitely higher or lower for any particular NSAID as compared to another (Ref).

Mechanism: Dose- and time-related; inhibition of COX-2 by NSAIDs results in a reduction in the production of prostaglandin I_2 (prostacyclin), a potent vasodilator and anticoagulant present in the vascular endothelium (Ref). Animal studies have shown that reduced prostacyclin activity may result in a predisposition to vascular injury (Ref). In addition, prostaglandins inhibit sodium resorption in the thick ascending loop of Henle and collecting tubule; therefore, a reduction in prostaglandin synthesis by NSAIDs may cause sodium and fluid retention and result in hypertension and decreased efficacy of diuretics (Ref).

Onset: Varied; increased risk is associated with both short- and long-term use and may be apparent within the first weeks following initiation of treatment (Ref); longer duration of therapy may further increase risk (Ref). Treatment with ketorolac is limited to short-term use (ie, 5 days).

Risk factors:

• ≥65 years of age

• Higher doses (especially with regards to CV thrombotic risk [(Ref)]

• Longer duration of use and frequent use (eg, ≥22 days per month (Ref)

- **Note**: Treatment with ketorolac is limited to short-term use (ie, 5 days).

• Preexisting CVD, including heart failure, or presence of risk factors for CVD (Ref)

- **Note**: Relative risk appears to be similar in those with and without known CVD or risk factors for CVD; however, absolute incidence of serious CV thrombotic events appears to be higher in patients with known CVD or risk factors for CVD due to an increased baseline risk (Ref).

- **Note**: The use of NSAIDs is contraindicated in the setting of CABG surgery due to an increased risk of MI and stroke in this patient population. Yet, retrospective studies specific to ketorolac have not shown an increased risk of death, MI, or bleeding events following use in postoperative CABG patients (Ref).

Gastrointestinal effects

Use of nonsteroidal anti-inflammatory drugs (NSAIDs), especially nonselective NSAIDs, such as ketorolac, is associated with an increased risk of serious gastrointestinal (GI) adverse events, including gastrointestinal inflammation, **gastrointestinal hemorrhage**, **gastrointestinal ulcer**, and **gastrointestinal perforation**; severity may range from asymptomatic to fatal (Ref). The risk of upper GI hemorrhage and/or perforation has been reported to be higher with ketorolac as compared to other nonselective NSAIDs (Ref).

Mechanism: Dose- and time-related; inhibition of cyclooxygenase (COX)-1 by NSAIDs results in a reduction in the production of mucosal-protective prostaglandin E_2 (Ref).

Onset: Varied; GI events can occur at any time during use and without warning symptoms. A longer duration of use (eg, ≥7 days (Ref)) is associated with a greater risk. Treatment with ketorolac is limited to short-term use (ie, 5 days).

Risk factors:

• Older age (eg, ≥65 years of age (Ref))

• Longer duration of use (eg, ≥5 days (Ref))

- **Note:** Treatment with ketorolac is limited to short-term use (ie, 5 days).

• Higher doses

• Prior history of peptic ulcer disease and/or GI bleeding (Ref)

• Concomitant use of agents known to increase the risk of GI bleeding (eg, aspirin (Ref), anticoagulants, corticosteroids (Ref), selective serotonin reuptake inhibitors (Ref))

• Comorbid *Helicobacter pylori* infection (Ref)

• Advanced liver disease/cirrhosis

• Coagulopathy

• Smoking

• Consumption of alcohol

• People with poor general health status

• Small intestine damage: Small intestine bacterial overgrowth (SIBO), including SIBO induced by proton pump inhibitor therapy, may be associated with an increased risk of small intestine damage (Ref)

Hematologic effects

Use of nonsteroidal anti-inflammatory drugs (NSAIDs), including ketorolac, is associated with **prolonged bleeding time** and an increased risk for hemorrhage, particularly gastrointestinal bleeding (Ref). Some studies have suggested that ketorolac is not associated with increased bleeding in the perioperative period (Ref).

In addition, drug-induced **hemolytic anemia** may occur with the use of NSAIDs (Ref). Rarely, NSAID use has been associated with potentially severe blood dyscrasias (eg, **agranulocytosis**, **aplastic anemia**, neutropenia, **thrombocytopenia**) (Ref).

Mechanism:

Prolonged bleeding time: Inhibition of cyclooxygenase (COX)-1 by nonselective NSAIDs causes a decrease in the production of prostaglandins, prostacyclins, and thromboxanes, including thromboxane A_2 (TxA_2) (Ref). As a result, patients may exhibit a decrease in platelet adhesion and aggregation and subsequent prolonged bleeding time (Ref).

Blood dyscrasias: Not clearly established; anemia may be due to occult or gross blood loss, fluid retention, or an incompletely described effect on erythropoiesis. May also be attributed to a drug-dependent immune mechanism (Ref).

Onset: Prolonged bleeding time: Rapid; suppression of platelet COX-1 activity occurs within hours of administration of an NSAID (Ref). Bleeding time was prolonged within hours of administration of a single dose of ketorolac in healthy volunteers (Ref); postoperatively, bleeding has been reported after a single dose of ketorolac (Ref).

Risk factors:

• Bleeding events:

- Preexisting coagulation disorders

- Concomitant use of agents known to increase the risk of bleeding (eg, anticoagulants (Ref), antithrombotics (Ref), antiplatelet agents [eg, aspirin, P2Y12 inhibitors], selective serotonin reuptake inhibitors (Ref), or serotonin norepinephrine reuptake inhibitors)

- Use during and immediately following surgical procedures (Ref)

• **Note:** Some studies have suggested that ketorolac is not associated with increased bleeding in the perioperative period (Ref)

- Age <21 days and <37 weeks corrected gestational age (Ref)

Hepatic effects

Nonsteroidal anti-inflammatory drugs (NSAIDs), including ketorolac, may cause mild transaminase elevations, especially with higher doses; one study found no association between use of parenteral ketorolac and hepatoxic effects (Ref). Rarely, serious liver injury may occur following use of NSAIDs (Ref). Cholestatic and mixed patterns of injury have been reported with the use of NSAIDs. NSAID-induced hepatic effects may occur following severe hypersensitivity reactions (eg, toxic epidermal necrosis, Stevens-Johnson syndrome) and are characterized by immune-mediated symptoms (eg, fever rash, eosinophilia, lymphadenopathy) (Ref). Severe liver injury requiring liver transplantation has also been reported following use of NSAIDs (Ref). Most cases of liver injury are likely reversible following discontinuation; full recovery may take several months (Ref). However, chronic vanishing bile duct syndrome with chronic liver failure has been reported following cholestatic liver injury secondary to ibuprofen (Ref).

Mechanism: Not clearly established; dose-related has been suggested (Ref). Proposed mechanisms include a toxic metabolite or a hypersensitivity reaction (Ref).

Onset: Varied; onset of NSAID-induced hepatotoxicity is generally classified as moderate (30 to 90 days) to long (>90 days) (Ref). For ibuprofen, a mean time of 12 days (range: 1 to 42 days) was reported in one study (Ref). Treatment with ketorolac is limited to short-term use (ie, 5 days).

Risk factors:

• Higher doses (Ref)

• Prior NSAID-related liver injury (Ref)

- **Note**: Cross-reactivity may occur between groups of NSAIDs (eg, cross-reactivity between propionic acid derivatives [ibuprofen, naproxen, ketoprofen]) (Ref). Ketorolac is a member of the pyrrolopyrrole group of NSAIDs.

Hypersensitivity reactions (immediate and delayed)

Hypersensitivity reactions (immediate and delayed) involving the skin (eg, **angioedema**, **urticaria**), airways (eg, dyspnea, rhinorrhea), and/or other organs have been reported (Ref). Clinical phenotypes of nonsteroidal anti-inflammatory drug (NSAID) hypersensitivity reactions include NSAID-exacerbated respiratory disease (NERD), NSAID-induced urticaria/angioedema (NIUA), NSAID-exacerbated cutaneous disease (NECD), and single NSAID-induced urticaria/angioedema or anaphylaxis (Ref). Delayed hypersensitivity reactions, including **drug rash with eosinophilia and systemic symptoms** (DRESS), and **Stevens-Johnson syndrome** (SJS)/**toxic epidermal necrolysis** (TEN) have also been associated with ketorolac.

Mechanism:

Immediate reactions: Non–dose-related; most reactions (ie, NERD, NECD, NIUA) are nonimmunologic related to inhibition of cyclooxygenase (COX)-1 with subsequent activation of mast cells and eosinophils causing release of inflammatory mediators including cysteinyl-leukotrienes (Ref). Some immediate reactions may be IgE mediated (Ref).

Delayed reactions: Delayed hypersensitivity reactions are T-cell mediated (Ref).

Onset:

Immediate reactions: Rapid; generally occurs within 1 hour of administration but may occur several hours after exposure (Ref).

Delayed reactions (including DRESS and SJS/TEN): Varied; generally occurs after 1 to 8 weeks after initiation (Ref).

Risk factors:

• Presence of chronic rhinosinusitis with nasal polyps, family history of NERD, and/or severe asthma may increase the risk of NERD (Ref). The prevalence of NERD in adult patients with asthma is ~10% to 20% (Ref).

• Chronic urticaria increases the risk of NECD (Ref). NSAID-induced reactions are less frequent and less intense when chronic urticaria is in remission or under control (Ref). Approximately 12% to 30% of patients with chronic idiopathic urticaria develop exacerbations of their disease with use of ketorolac and other COX-1 inhibitors (Ref).

• Cross-reactivity between aspirin and NSAIDs, including ketorolac (with predominant COX-1 inhibition) have been described in patients with a history of NERD, NECD, and NIUA (Ref). Cross-reactivity between aspirin/NSAID and acetaminophen, a weak COX inhibitor, and between aspirin/NSAID and nonselective COX-2 inhibitors (eg, meloxicam, nimesulide) may occur (Ref). Although selective COX-2 inhibitors (eg, celecoxib, etoricoxib) are generally tolerated in patients with NERD (Ref), cross-reactions may occur, especially in patients with histories of urticaria/angioedema (Ref).

Kidney effects

Hemodynamically mediated **acute kidney injury** (AKI) may occur following use of either cyclooxygenase (COX)-2 selective nonsteroidal anti-inflammatory drugs (NSAIDs) (ie, coxibs) or nonselective NSAIDs, including ketorolac (Ref); the risk may be greater with nonselective NSAIDs (Ref). The risk of developing AKI is decreased upon discontinuation (Ref). In patients who develop AKI, kidney function is likely to return to baseline following prompt discontinuation of the offending NSAID and supportive care (Ref); however, the mechanism of the damage and other concurrent factors can contribute to irreversibility.

Mechanism: Dose- and time-related; inhibition of COX-1 and COX-2 by NSAIDs results in a reduced production of nephroprotective prostaglandins and subsequent attenuation of renal vasodilation (Ref). In addition, an increase in vasoconstriction of the afferent arteriole and impaired renal blood flow causes a reduction in the glomerular capillary pressure and filtration (Ref).

Onset: Rapid; may occur within hour to days of treatment initiation (Ref).

Risk factors:

• AKI:

- Preexisting kidney impairment

- Chronic kidney disease

- ≥65 years of age (Ref)

• **Note:** NSAID-associated AKI may also occur in pediatric patients, even at therapeutic doses (Ref).

• Hemodynamically mediated AKI:

- Preexisting conditions that result in decreased effective arterial circulation (ie, conditions where renal blood flow/renal perfusion may be dependent on prostaglandin-mediated vasodilation) (Ref):

• Volume depletion (eg, due to concomitant diuretic use, vomiting)

• Heart failure (Ref)

• Cirrhosis and ascites (Ref)

• Nephrotic syndrome

- Concomitant use of diuretics, angiotensin-converting enzyme inhibitors, angiotensin receptor blockers, or calcineurin inhibitors (Ref)

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Some reactions listed are based on reports for other agents in this same pharmacologic class and may not be specifically reported for ketorolac.

>10%:

Gastrointestinal: Abdominal pain, dyspepsia, nausea

Hepatic: Increased liver enzymes (≤15%)

Nervous system: Headache

1% to 10%:

Cardiovascular: Edema, hypertension

Dermatologic: Diaphoresis, pruritus, skin rash

Gastrointestinal: Constipation, diarrhea, flatulence, gastrointestinal fullness, gastrointestinal hemorrhage, gastrointestinal perforation, gastrointestinal ulcer, heartburn, stomatitis, vomiting

Hematologic & oncologic: Anemia, prolonged bleeding time, purpuric disease

Local: Pain at injection site (IM, IV)

Nervous system: Dizziness, drowsiness

Otic: Tinnitus

Renal: Altered kidney function

<1%:

Cardiovascular: Heart failure, palpitations, syncope, tachycardia

Dermatologic: Alopecia, ecchymoses, pallor, skin photosensitivity, urticaria

Endocrine & metabolic: Increased thirst, weight changes

Gastrointestinal: Anorexia, dysgeusia, eructation, esophagitis, gastritis, glossitis, hematemesis, increased appetite, melena, rectal hemorrhage, xerostomia

Genitourinary: Cystitis, dysuria, exacerbation of urinary frequency, hematuria, infertility, oliguria, proteinuria, urinary retention

Hematologic & oncologic: Eosinophilia, leukopenia, thrombocytopenia

Hepatic: Hepatitis, jaundice

Infection: Infection, sepsis

Nervous system: Abnormal dreams, anxiety, asthenia, changes in thinking, confusion, depression, euphoria, extrapyramidal reaction, hallucination, insomnia, lack of concentration, malaise, nervousness, paresthesia, stupor, tremor, vertigo

Neuromuscular & skeletal: Hyperkinetic muscle activity

Ophthalmic: Blurred vision, visual disturbance

Otic: Hearing loss

Renal: Interstitial nephritis, kidney failure, polyuria

Respiratory: Asthma, cough, dyspnea, epistaxis, pulmonary edema, rhinitis

Miscellaneous: Fever

Frequency not defined:

Cardiovascular: Coronary thrombosis

Gastrointestinal: Peptic ulcer

Nervous system: Cerebrovascular accident

Postmarketing:

Cardiovascular: Acute myocardial infarction, bradycardia, cardiac arrhythmia, chest pain, flushing, hypotension, vasculitis

Dermatologic: Bullous skin disease, erythema multiforme, exfoliative dermatitis, fixed drug eruption, Stevens-Johnson syndrome, toxic epidermal necrolysis

Endocrine & metabolic: Hyperglycemia, hyperkalemia (Haragsim 1994), hyponatremia

Gastrointestinal: Acute pancreatitis (Famularo 2002), aphthous stomatitis, exacerbation of Crohn disease, exacerbation of ulcerative colitis

Genitourinary: Azotemia

Hematologic: Agranulocytosis, aplastic anemia, hemolytic anemia, hemolytic-uremic syndrome (Randi 1993), lymphadenopathy, pancytopenia, postoperative hematoma, postoperative wound bleeding

Hepatic: Hepatic failure, hepatotoxicity (idiosyncratic) (Chalasani 2021)

Hypersensitivity: Anaphylaxis (Yousefi 2020), angioedema (Shapiro 1994), drug reaction with eosinophilia and systemic symptoms, hypersensitivity reaction, nonimmune anaphylaxis (Goetz 1992), tongue edema

Nervous system: Aseptic meningitis (Shahin 2024), coma, psychosis, seizure

Neuromuscular & skeletal: Myalgia

Ophthalmic: Conjunctivitis

Renal: Acute kidney injury (Haragsim 1994), flank pain, nephrotic syndrome, renal papillary necrosis (Witting 1996)

Respiratory: Bronchospasm (Chen 1994), laryngeal edema, pneumonia, respiratory depression

Contraindications

Hypersensitivity to ketorolac, aspirin, other nonsteroidal anti-inflammatory drugs (NSAIDs), or any component of the formulation; active or history of peptic ulcer disease; recent or history of GI bleeding or perforation; history of asthma, urticaria, or allergic-type reactions after taking aspirin or other NSAIDs; advanced renal disease or patients at risk for renal failure due to volume depletion; prophylactic analgesic before any major surgery; suspected or confirmed cerebrovascular bleeding, hemorrhagic diathesis, incomplete hemostasis, or high risk of bleeding; concurrent use with aspirin, other NSAIDs, probenecid, or pentoxifylline; epidural or intrathecal administration (injection only); use in the setting of coronary artery bypass graft surgery; labor and delivery.

Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Canadian labeling: Additional contraindications (not in US labeling): Intraoperative use; coagulation disorders; active GI bleeding; postoperative patients with high-bleeding risk; severe uncontrolled heart failure; inflammatory bowel disease; severe hepatic impairment or active hepatic disease; moderate to severe renal impairment (serum creatinine >442 micromol/L and/or creatinine clearance <30 mL/minute) or deteriorating renal disease; known hyperkalemia; third trimester of pregnancy; breast-feeding; use in children and adolescents <18 years of age

Warnings/Precautions

Concerns related to adverse effects:

• Hyperkalemia: Nonsteroidal anti-inflammatory drug (NSAID) use may increase the risk of hyperkalemia, particularly in patients ≥65 years of age, in patients with diabetes or renal disease, and with concomitant use of other agents capable of inducing hyperkalemia (eg, angiotensin-converting enzyme inhibitors). Monitor potassium closely.

Disease-related concerns:

• Aseptic meningitis: May increase the risk of aseptic meningitis, especially in patients with systemic lupus erythematosus and mixed connective tissue disorders.

• Bariatric surgery: Gastric ulceration: Avoid chronic use of oral nonselective NSAIDs after bariatric surgery; development of anastomotic ulcerations/perforations may occur (Bhangu 2014; Mechanick 2020). Short-term use of celecoxib or IV ketorolac are recommended as part of a multimodal pain management strategy for postoperative pain (Chou 2016; Horsley 2019; Thorell 2016).

• Hepatic impairment: Avoid use; patients with advanced hepatic disease are at an increased risk of GI bleeding and kidney failure with NSAIDs (AASLD [Biggins 2021]; AASLD [Runyon 2013]).

Special populations:

• Older adult: Patients ≥65 years of age are at greater risk for serious GI, cardiovascular, and/or renal adverse events; use with caution.

Other warnings/precautions:

• Surgical/Dental procedures: Withhold for at least 4 to 6 half-lives prior to surgical or dental procedures.

Warnings: Additional Pediatric Considerations

In neonates, bleeding events have been reported; in a retrospective analysis of 57 postsurgical neonates and infants (age range: 0 to 3 months), 17.2% of patients experienced a bleeding event, most were PNA <21 days and those with PNA <14 days had a significantly higher risk than older neonates (Aldrink 2011). Acute kidney injury (AKI) has been observed in pediatric patients; with ketorolac use following cardiothoracic surgery, an increased risk of AKI was observed in neonates and infants who underwent a bidirectional Glenn procedure and were receiving concomitant aspirin therapy (Moffett 2013).

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Injection, as tromethamine:

Generic: 15 mg/mL (1 mL); 30 mg/mL (1 mL)

Solution, Injection, as tromethamine [preservative free]:

Generic: 15 mg/mL (1 mL); 30 mg/mL (1 mL)

Solution, Intramuscular, as tromethamine:

Generic: 60 mg/2 mL (2 mL)

Solution, Intramuscular, as tromethamine [preservative free]:

Generic: 60 mg/2 mL (2 mL)

Tablet, Oral, as tromethamine:

Generic: 10 mg

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Ketorolac Tromethamine Injection)

15 mg/mL (per mL): $0.84 - $4.50

30 mg/mL (per mL): $0.84 - $7.84

Solution (Ketorolac Tromethamine Intramuscular)

60 mg/2 mL (per mL): $0.54 - $4.92

Tablets (Ketorolac Tromethamine Oral)

10 mg (per each): $1.10 - $2.27

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intramuscular:

Toradol: 10 mg/mL (1 mL) [contains alcohol, usp]

Solution, Intramuscular, as tromethamine:

Generic: 30 mg/mL (1 mL)

Tablet, Oral, as tromethamine:

Toradol: 10 mg

Generic: 10 mg

Administration: Adult

Oral: May administer with food to reduce GI upset.

IM: Administer slowly and deeply into the muscle.

IV: Administer IV bolus over a minimum of 15 seconds.

Administration: Pediatric

Oral: Tablet: May administer with food or milk to decrease GI upset.

Parenteral:

IM: Administer slowly and deeply into muscle; 60 mg per 2 mL vial is for IM use only.

IV: Administer undiluted over at least 15 seconds; in children, ketorolac has been infused over 1 to 5 minutes (Ref).

Storage/Stability

Injection: Store at 20°C to 25°C (68°F to 77°F); excursions permitted to 15°C to 30°C (59°F to 86°F). Protect from light. Injection is clear and has a slight yellow color. Precipitation may occur at relatively low pH values.

Tablet: Store at 20°C to 25°C (68°F to 77°F). Protect from light and excessive humidity.

Use: Labeled Indications

Pain management, acute: Short-term (≤5 days) management of acute pain.

Use: Off-Label: Adult

Migraine, severe, acute treatment

Medication Safety Issues

Sound-alike/look-alike issues:

Ketorolac may be confused with Ketalar, ketamine, methadone

Toradol may be confused with Foradil, Inderal, Tegretol, traMADol, tromethamine

Older Adult: High-Risk Medication:

Beers Criteria: Ketorolac is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) due to an increased risk of GI bleeding, peptic ulcer disease, and acute kidney injury (Beers Criteria [AGS 2023]).

NSAIDs are identified in the Screening Tool of Older Person's Prescriptions (STOPP) criteria as potentially inappropriate medications in older adults (≥65 years of age) for long-term use in the treatment of gout or osteoarthritis where other agents have not been tried (eg, acetaminophen for osteoarthritis, xanthine-oxidase inhibitors for gout). In addition, some disease states of concern include heart failure, renal impairment, peptic ulcer disease, gastrointestinal bleeding, severe hypertension, and coronary, cerebral or peripheral vascular disease (O'Mahony 2023).

International issues:

Toradol [Canada and multiple international markets] may be confused with Theradol brand name for tramadol [Netherlands]

Metabolism/Transport Effects

Substrate of OAT1/3

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

5-Aminosalicylic Acid Derivatives: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of 5-Aminosalicylic Acid Derivatives. *Risk C: Monitor*

Abciximab: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Abrocitinib: Agents with Antiplatelet Effects may increase antiplatelet effects of Abrocitinib. *Risk X: Avoid*

Acalabrutinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Acemetacin: May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk X: Avoid*

Alcohol (Ethyl): May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of GI bleeding may be increased with this combination. *Risk C: Monitor*

Aliskiren: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Aliskiren. Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Aliskiren. *Risk C: Monitor*

Aminoglycosides: Nonsteroidal Anti-Inflammatory Agents may decrease excretion of Aminoglycosides. Data only in premature infants. *Risk C: Monitor*

Aminolevulinic Acid (Systemic): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Systemic). *Risk X: Avoid*

Aminolevulinic Acid (Topical): Photosensitizing Agents may increase photosensitizing effects of Aminolevulinic Acid (Topical). *Risk C: Monitor*

Anagrelide: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Angiotensin II Receptor Blockers: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Angiotensin II Receptor Blockers. The combination of these two agents may also significantly decrease glomerular filtration and renal function. Angiotensin II Receptor Blockers may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the combination may result in a significant decrease in renal function. *Risk C: Monitor*

Angiotensin-Converting Enzyme Inhibitors: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Angiotensin-Converting Enzyme Inhibitors. Angiotensin-Converting Enzyme Inhibitors may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the combination may result in a significant decrease in renal function. *Risk C: Monitor*

Anticoagulants (Miscellaneous Agents): Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Anticoagulants (Miscellaneous Agents). *Risk C: Monitor*

Antiplatelet Agents (P2Y12 Inhibitors): Agents with Antiplatelet Effects may increase antiplatelet effects of Antiplatelet Agents (P2Y12 Inhibitors). *Risk C: Monitor*

Aspirin: Ketorolac (Systemic) may increase adverse/toxic effects of Aspirin. An increased risk of bleeding may be associated with use of this combination. Ketorolac (Systemic) may decrease cardioprotective effects of Aspirin. *Risk X: Avoid*

Bemiparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Bemiparin. Management: Avoid this combination if possible, due to an increased risk of bleeding. If coadministration cannot be avoided, monitor patients closely for clinical and laboratory evidence of bleeding. *Risk D: Consider Therapy Modification*

Beta-Blockers: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Beta-Blockers. *Risk C: Monitor*

Bile Acid Sequestrants: May decrease absorption of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Bisphosphonate Derivatives: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Bisphosphonate Derivatives. Both an increased risk of gastrointestinal ulceration and an increased risk of nephrotoxicity are of concern. *Risk C: Monitor*

Caplacizumab: Agents with Antiplatelet Effects may increase adverse/toxic effects of Caplacizumab. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Cardiac Glycosides: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Cardiac Glycosides. *Risk C: Monitor*

Collagenase (Systemic): Agents with Antiplatelet Effects may increase adverse/toxic effects of Collagenase (Systemic). Specifically, the risk of injection site bruising and or bleeding may be increased. *Risk C: Monitor*

Corticosteroids (Systemic): May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). *Risk C: Monitor*

CycloSPORINE (Systemic): Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of CycloSPORINE (Systemic). Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of CycloSPORINE (Systemic). CycloSPORINE (Systemic) may increase serum concentrations of Nonsteroidal Anti-Inflammatory Agents. Management: Consider alternatives to nonsteroidal anti-inflammatory agents (NSAIDs). Monitor for evidence of nephrotoxicity, as well as increased serum cyclosporine concentrations and systemic effects (eg, hypertension) during concomitant therapy with NSAIDs. *Risk D: Consider Therapy Modification*

Dasatinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Deferasirox: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Deferasirox. Specifically, the risk for GI ulceration/irritation or GI bleeding may be increased. *Risk C: Monitor*

Deoxycholic Acid: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Desmopressin: Nonsteroidal Anti-Inflammatory Agents may increase hyponatremic effects of Desmopressin. *Risk C: Monitor*

Dichlorphenamide: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). Management: Consider alternatives to the coadministration of dichlorphenamide and OAT1/3 substrates when possible, as product labeling states the concomitant use of these agents is not recommended. If combined, monitor for adverse effects of these OAT1/3 substrates. *Risk D: Consider Therapy Modification*

Direct Oral Anticoagulants (DOACs): Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Direct Oral Anticoagulants (DOACs). *Risk C: Monitor*

Drospirenone-Containing Products: May increase hyperkalemic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Enoxaparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Enoxaparin. Management: Discontinue nonselective NSAIDs prior to initiation of enoxaparin whenever possible. If coadministration cannot be avoided, monitor patients closely for clinical and laboratory evidence of bleeding. *Risk D: Consider Therapy Modification*

Eplerenone: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Eplerenone. Nonsteroidal Anti-Inflammatory Agents may increase hyperkalemic effects of Eplerenone. *Risk C: Monitor*

Fexinidazole: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). Management: Avoid use of fexinidazole with OAT1/3 substrates when possible. If combined, monitor for increased OAT1/3 substrate toxicities. *Risk D: Consider Therapy Modification*

Fondaparinux: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Fondaparinux. Management: Discontinue nonselective nonsteroidal anti-inflammatory agents prior to the initiation of fondaparinux, if possible. If coadministration is required, monitor patients closely for signs and symptoms of bleeding. *Risk D: Consider Therapy Modification*

Glycoprotein IIb/IIIa Inhibitors: Agents with Antiplatelet Effects may increase antiplatelet effects of Glycoprotein IIb/IIIa Inhibitors. *Risk C: Monitor*

Heparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Heparin. *Risk C: Monitor*

Heparins (Low Molecular Weight): Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Heparins (Low Molecular Weight). *Risk C: Monitor*

Herbal Products with Anticoagulant/Antiplatelet Effects: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

HydrALAZINE: Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of HydrALAZINE. *Risk C: Monitor*

Ibritumomab Tiuxetan: Agents with Antiplatelet Effects may increase antiplatelet effects of Ibritumomab Tiuxetan. *Risk C: Monitor*

Ibrutinib: Agents with Antiplatelet Effects may increase adverse/toxic effects of Ibrutinib. Specifically, the risk of bleeding and hemorrhage may be increased. *Risk C: Monitor*

Inotersen: Agents with Antiplatelet Effects may increase adverse/toxic effects of Inotersen. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Ketorolac (Nasal): May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk X: Avoid*

Leflunomide: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Limaprost: May increase adverse/toxic effects of Agents with Antiplatelet Effects. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Lithium: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Lithium. *Risk C: Monitor*

Loop Diuretics: Nonsteroidal Anti-Inflammatory Agents may decrease diuretic effects of Loop Diuretics. Loop Diuretics may increase nephrotoxic effects of Nonsteroidal Anti-Inflammatory Agents. Management: Monitor for evidence of kidney injury or decreased therapeutic effects of loop diuretics with concurrent use of an NSAID. Consider avoiding concurrent use in CHF or cirrhosis. Concomitant use of bumetanide with indomethacin is not recommended. *Risk D: Consider Therapy Modification*

Macimorelin: Coadministration of Nonsteroidal Anti-Inflammatory Agents and Macimorelin may alter diagnostic results. *Risk X: Avoid*

MetFORMIN: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of MetFORMIN. *Risk C: Monitor*

Methotrexate: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Methotrexate. Management: Avoid coadministration of higher dose methotrexate (such as that used for the treatment of oncologic conditions) and NSAIDs. Use caution if coadministering lower dose methotrexate and NSAIDs. *Risk D: Consider Therapy Modification*

Methoxsalen (Systemic): Photosensitizing Agents may increase photosensitizing effects of Methoxsalen (Systemic). *Risk C: Monitor*

Methoxyflurane: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Methoxyflurane. *Risk X: Avoid*

Mifamurtide: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Mifamurtide. *Risk X: Avoid*

Miscellaneous Antiplatelets: Agents with Antiplatelet Effects may increase antiplatelet effects of Miscellaneous Antiplatelets. *Risk C: Monitor*

Nadroparin: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Nadroparin. Management: Coadministration of NSAIDs and nadroparin is not recommended due to an increased risk of bleeding. If coadministration is required, monitor patients closely for clinical and laboratory signs of bleeding. *Risk D: Consider Therapy Modification*

Naftazone: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Neuromuscular-Blocking Agents (Nondepolarizing): Ketorolac (Systemic) may increase adverse/toxic effects of Neuromuscular-Blocking Agents (Nondepolarizing). Specifically, episodes of apnea have been reported in patients using this combination. *Risk C: Monitor*

Nitisinone: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Nonsteroidal Anti-Inflammatory Agents (Topical): Nonsteroidal Anti-Inflammatory Agents (Topical) may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of gastrointestinal (GI) toxicity is increased. Management: Coadministration of systemic nonsteroidal anti-inflammatory drugs (NSAIDs) and topical NSAIDs is not recommended. If systemic NSAIDs and topical NSAIDs, ensure the benefits outweigh the risks and monitor for increased NSAID toxicities. *Risk D: Consider Therapy Modification*

Nonsteroidal Anti-Inflammatory Agents: May increase adverse/toxic effects of Ketorolac (Systemic). *Risk X: Avoid*

Obinutuzumab: Agents with Antiplatelet Effects may increase adverse/toxic effects of Obinutuzumab. Specifically, the risk of bleeding may be increased. Management: Consider avoiding coadministration of obinutuzumab and agents with antiplatelet effects, especially during the first cycle of obinutuzumab therapy. *Risk D: Consider Therapy Modification*

Omacetaxine: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Omacetaxine. Specifically, the risk for bleeding-related events may be increased. *Risk C: Monitor*

Pentosan Polysulfate Sodium: Agents with Antiplatelet Effects may increase adverse/toxic effects of Pentosan Polysulfate Sodium. Specifically, the risk of hemorrhage may be increased. *Risk C: Monitor*

Pentoxifylline: Ketorolac (Systemic) may increase adverse/toxic effects of Pentoxifylline. Specifically, the risk of bleeding may be increased with this combination. *Risk X: Avoid*

Phenylbutazone: May increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk X: Avoid*

Pirtobrutinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Polyethylene Glycol-Electrolyte Solution: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Polyethylene Glycol-Electrolyte Solution. *Risk C: Monitor*

Porfimer: Photosensitizing Agents may increase photosensitizing effects of Porfimer. *Risk X: Avoid*

Potassium Salts: Nonsteroidal Anti-Inflammatory Agents may increase hyperkalemic effects of Potassium Salts. *Risk C: Monitor*

Potassium-Sparing Diuretics: Nonsteroidal Anti-Inflammatory Agents may increase hyperkalemic effects of Potassium-Sparing Diuretics. Nonsteroidal Anti-Inflammatory Agents may decrease antihypertensive effects of Potassium-Sparing Diuretics. *Risk C: Monitor*

PRALAtrexate: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of PRALAtrexate. More specifically, NSAIDS may decrease the renal excretion of pralatrexate. Management: Avoid coadministration of pralatrexate with nonsteroidal anti-inflammatory drugs (NSAIDs). If coadministration cannot be avoided, closely monitor for increased pralatrexate serum levels or toxicity. *Risk D: Consider Therapy Modification*

Pretomanid: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Probenecid: May increase serum concentrations of Ketorolac (Systemic). *Risk X: Avoid*

Prostaglandins (Ophthalmic): Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Prostaglandins (Ophthalmic). Nonsteroidal Anti-Inflammatory Agents may also enhance the therapeutic effects of Prostaglandins (Ophthalmic). *Risk C: Monitor*

Quinolones: Nonsteroidal Anti-Inflammatory Agents may increase neuroexcitatory and/or seizure-potentiating effects of Quinolones. Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Quinolones. *Risk C: Monitor*

Salicylates: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase adverse/toxic effects of Salicylates. An increased risk of bleeding may be associated with use of this combination. Management: Consider alternatives when possible, as product labeling states the concomitant use of these agents is not recommended. If combined, monitor for gastrointestinal toxicity (eg, gastrointestinal bleeding, ulceration, perforation). *Risk D: Consider Therapy Modification*

Selective Serotonin Reuptake Inhibitor: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). Nonsteroidal Anti-Inflammatory Agents (Nonselective) may decrease therapeutic effects of Selective Serotonin Reuptake Inhibitor. Management: Consider alternatives to NSAIDs. Monitor for evidence of bleeding and diminished antidepressant effects. It is unclear whether COX-2-selective NSAIDs reduce risk. *Risk D: Consider Therapy Modification*

Selumetinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Serotonin/Norepinephrine Reuptake Inhibitor: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents (Nonselective). *Risk C: Monitor*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Sodium Phosphates: May increase nephrotoxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of acute phosphate nephropathy may be enhanced. *Risk C: Monitor*

Sulprostone: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Sulprostone. *Risk X: Avoid*

Tacrolimus (Systemic): Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Tacrolimus (Systemic). *Risk C: Monitor*

Taurursodiol: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk X: Avoid*

Temoporfin: Photosensitizing Agents may increase photosensitizing effects of Temoporfin. *Risk X: Avoid*

Tenofovir Products: Nonsteroidal Anti-Inflammatory Agents may increase nephrotoxic effects of Tenofovir Products. Management: Seek alternatives to these combinations whenever possible. Avoid use of tenofovir with multiple NSAIDs or any NSAID given at a high dose due to a potential risk of acute renal failure. Diclofenac appears to confer the most risk. *Risk D: Consider Therapy Modification*

Teriflunomide: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Thiazide and Thiazide-Like Diuretics: Nonsteroidal Anti-Inflammatory Agents may decrease therapeutic effects of Thiazide and Thiazide-Like Diuretics. Thiazide and Thiazide-Like Diuretics may increase nephrotoxic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Thrombolytic Agents: Agents with Antiplatelet Effects may increase adverse/toxic effects of Thrombolytic Agents. Specifically, the risk of bleeding may be increased. *Risk C: Monitor*

Tipranavir: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Tolperisone: Nonsteroidal Anti-Inflammatory Agents may increase adverse/toxic effects of Tolperisone. Specifically, the risk of hypersensitivity reactions may be increased. Tolperisone may increase therapeutic effects of Nonsteroidal Anti-Inflammatory Agents. *Risk C: Monitor*

Tricyclic Antidepressants: May increase antiplatelet effects of Nonsteroidal Anti-Inflammatory Agents. Tricyclic Antidepressants may increase adverse/toxic effects of Nonsteroidal Anti-Inflammatory Agents. Specifically, the risk of major adverse cardiac events (MACE), hemorrhagic stroke, ischemic stroke, and heart failure may be increased. *Risk C: Monitor*

Vaborbactam: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Vadadustat: May increase serum concentrations of OAT1/3 Substrates (Clinically Relevant). *Risk C: Monitor*

Vancomycin: Nonsteroidal Anti-Inflammatory Agents may increase serum concentrations of Vancomycin. *Risk C: Monitor*

Verteporfin: Photosensitizing Agents may increase photosensitizing effects of Verteporfin. *Risk C: Monitor*

Vitamin E (Systemic): May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Vitamin K Antagonists: Nonsteroidal Anti-Inflammatory Agents (Nonselective) may increase anticoagulant effects of Vitamin K Antagonists. *Risk C: Monitor*

Volanesorsen: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Zanubrutinib: May increase antiplatelet effects of Agents with Antiplatelet Effects. *Risk C: Monitor*

Food Interactions

High-fat meals may delay time to peak (by ~1 hour) and decrease peak concentrations. Management: Administer tablet with food or milk to decrease gastrointestinal distress.

Reproductive Considerations

Nonsteroidal anti-inflammatory drugs may delay or prevent rupture of ovarian follicles. This may be associated with infertility that is reversible upon discontinuation of the medication. Consider discontinuing use in patients having difficulty conceiving or those undergoing investigation of fertility.

Pregnancy Considerations

Ketorolac crosses the placenta (Walker 1988).

The use of nonsteroidal anti-inflammatory drugs (NSAIDs) close to conception may be associated with an increased risk of miscarriage due to cyclooxygenase-2 inhibition interfering with implantation (Bermas 2014; Bloor 2013).

Birth defects have been observed following in utero NSAID exposure in some studies; however, data are conflicting (Bloor 2013). Nonteratogenic effects, including prenatal constriction of the ductus arteriosus, persistent pulmonary hypertension of the newborn, oligohydramnios, necrotizing enterocolitis, renal dysfunction or failure, and intracranial hemorrhage, have been observed in the fetus/neonate following in utero NSAID exposure (Bermas 2014; Bloor 2013). Maternal NSAID use may cause fetal renal dysfunction leading to oligohydramnios. Although rare, this may occur as early as 20 weeks' gestation and is more likely to occur with prolonged maternal use. Oligohydramnios may be reversible following discontinuation of the NSAID (Dathe 2019; FDA 2020). In addition, nonclosure of the ductus arteriosus postnatally may occur and be resistant to medical management (Bermas 2014; Bloor 2013).

Maternal use of NSAIDs should be avoided beginning at ~20 weeks' gestation. If NSAID use is necessary between 20 and 30 weeks' gestation, limit use to the lowest effective dose and shortest duration possible; consider ultrasound monitoring of amniotic fluid if treatment extends beyond 48 hours and discontinue the NSAID if oligohydramnios is found (FDA 2020). Because NSAIDs may cause premature closure of the ductus arteriosus, prescribing information for ketorolac specifically states use should be avoided starting at ~30 weeks' gestation

Due to pregnancy-induced physiologic changes, some pharmacokinetic properties of ketorolac may be altered. Ketorolac has S and R enantiomers; pharmacologic activity is associated with S-ketorolac enantiomer. The clearance of S-ketorolac was found to increase at delivery compared to postpartum values; the peak serum concentration was decreased (Kulo 2017; Välitalo 2017).

The use of ketorolac in labor and delivery is contraindicated because it may adversely affect fetal circulation and inhibit uterine contractions. The risk of uterine hemorrhage may be increased. NSAIDs may be used as part of multimodal pain management following cesarean delivery (ACOG 2019; Hostage 2023; Murdoch 2023).

Breastfeeding Considerations

Ketorolac is present in breast milk (Wischnik 1989).

The distribution of ketorolac into breast milk was studied in 10 women who were 2 to 6 days postpartum. Patients were given ketorolac 10 mg orally 4 times/day for 2 days. Simultaneous breast milk and maternal serum samples were drawn 4 times over the dosing period and the following day (day 3). Ketorolac concentrations were below the limit of quantification (5 ng/mL) in 4 patients at every sampling time, in all patients when measured prior to and 6 hours after the morning dose on days 1 and 2, and in all patients on day 3. When measurable, breast milk concentrations of ketorolac ranged from 5.2 to 7.9 ng/mL. The M/P ratio, when able to be calculated, ranged from 0.015 to 0.037. Using a breast milk concentration of 7.9 ng/mL, an infant weight of 3 kg, infant intake of 1 L of milk/day, and a maternal weight of 60 kg, the authors of the study estimated the exposure of ketorolac to the breastfeeding infant to be 0.16% to 0.40% of the maternal dose (Wischnik 1989). In general, breastfeeding is considered acceptable when the relevant infant dose is <10% (Anderson 2016; Ito 2000).

Data related to breastfeeding are available from 3 studies that evaluated use of ketorolac as part of a multimodal protocol for pain following cesarean delivery. In 1 study, women were given a single dose of ketorolac. There were no differences observed between neonates exposed to the protocol containing ketorolac compared to routine standard care without ketorolac in relation to breastfeeding, neonatal growth, sedation and respiratory depression (Hadley 2019). In a second study, ketorolac was used as needed or as a scheduled dose every 6 hours for 24 hours. More mothers who received the scheduled doses exclusively breastfed their newborns; the duration of breastfeeding was also increased (Teigen 2020). In a third study, the maternal dose of ketorolac (15 or 30 mg) was not found to influence the incidence of breastfeeding when used for up to 24 hours postpartum (Yurashevich 2020).

Nonopioid analgesics, including nonsteroidal anti-inflammatory drugs (NSAIDs), are preferred for breastfeeding patients who require pain control peripartum or for surgery outside of the postpartum period (ABM [Martin 2018]; ABM [Reece-Stremtan 2017]).

The manufacturer recommends that caution be used if administered to patients who are breastfeeding. Maternal use of NSAIDs should be avoided if the breastfeeding infant has platelet dysfunction, thrombocytopenia, or a ductal-dependent cardiac lesion (ABM [Martin 2018]; ABM [Reece-Stremtan 2017]; Bloor 2013). Agents other than ketorolac are preferred in breastfeeding patients at risk of hemorrhage (ABM [Reece-Stremtan 2017]),

Dietary Considerations

Administer tablet with food or milk to decrease gastrointestinal distress.

Monitoring Parameters

Monitor response (pain, range of motion, grip strength, mobility, ADL function), inflammation; observe for weight gain, edema; renal function (serum creatinine, BUN, urine output); CBC and platelets, liver function; chemistry profile; blood pressure; observe for bleeding, bruising; evaluate GI effects (abdominal pain, bleeding, dyspepsia); mental confusion, disorientation; signs and symptoms of hypersensitivity (eg, fever, rash, lymphadenopathy, eosinophilia) in association with other organ system involvement (eg, hepatitis, nephritis, hematological abnormalities, myocarditis, myositis).

Reference Range

Serum concentration: Therapeutic: 0.3 to 5 mcg/mL; Toxic: >5 mcg/mL

Mechanism of Action

Reversibly inhibits cyclooxygenase-1 and 2 (COX-1 and 2) enzymes, which results in decreased formation of prostaglandin precursors; has antipyretic, analgesic, and anti-inflammatory properties

Other proposed mechanisms not fully elucidated (and possibly contributing to the anti-inflammatory effect to varying degrees), include inhibiting chemotaxis, altering lymphocyte activity, inhibiting neutrophil aggregation/activation, and decreasing proinflammatory cytokine levels.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Analgesic: ~30 minutes.

Peak effect: Analgesic: ~2 to 3 hours.

Duration: Analgesic: 4 to 6 hours.

Absorption: Oral: Well absorbed (100%), administration after a high-fat meal decreased peak and delayed time to peak concentrations by ~1 hour; IM: Rapid and complete.

Distribution: Poor penetration into cerebrospinal fluid.

Infants: V_{dss}: ~0.2 L/kg (McLay 2018).

Children and Adolescents <16 years: V_{dss}: 0.18 L/kg (McLay 2018).

Adults: V_d beta: Oral, IM: 0.17 ± 0.04 L/kg; IV: 0.21 ± 0.04 L/kg.

Protein binding: 99%.

Metabolism: Hepatic; undergoes hydroxylation and glucuronide conjugation.

Bioavailability: Oral, IM: 100%.

Half-life elimination:

Infants ≥6 months: S-enantiomer: 0.83 ± 0.7 hours; R-enantiomer: 4 ± 0.8 hours (Lynn 2007).

Children and Adolescents ≤16 years: 3 ± 1.1 hours (Dsida 2002).

Adults:

Mean: ~5 hours; Range: 2 to 9 hours [S-enantiomer ~2.5 hours (biologically active); R-enantiomer ~5 hours]; Prolonged 30% to 50% in elderly.

With renal impairment: S_{cr} 1.9 to 5 mg/dL: Mean: ~11 hours; Range: 4 to 19 hours.

Renal dialysis patients: Mean: ~14 hours; Range: 8 to 40 hours.

Time to peak, serum: Oral: ~45 minutes; IM: ~30 to 45 minutes; IV: 1 to 3 minutes.

Excretion: Urine (92%, ~60% as unchanged drug); feces ~6%.

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Clearance is reduced and half-life is increased in renal impairment. AUC is increased by ~100% and volume of distribution increases.

Pediatric: V_d and clearance are higher in pediatric patients compared to adults.

Older adult: Half-life is longer in patients ≥65 years of age.

Medication Guide and/or Vaccine Information Statement (VIS)

An FDA-approved patient medication guide, which is available with the product information and at NSAIDs Class Medication Guide, must be dispensed with this medication.

Patient Counseling Points

(For additional information see ”Ketorolac (systemic): Patient drug information”)

What is this drug used for?

• It is used to ease pain.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Headache

• Stomach pain or heartburn

• Upset stomach

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• Bleeding like throwing up or coughing up blood; vomit that looks like coffee grounds; blood in the urine; black, red, or tarry stools; bleeding from the gums; abnormal vaginal bleeding; bruises without a cause or that get bigger; or bleeding you cannot stop

• Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain

• High potassium levels like a heartbeat that does not feel normal; feeling confused; feeling weak, lightheaded, or dizzy; feeling like passing out; numbness or tingling; or shortness of breath

• High blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Chest pain or pressure

• Weakness on 1 side of the body, trouble speaking or thinking, change in balance, drooping on one side of the face, or blurred eyesight

• Shortness of breath, a big weight gain, or swelling in the arms or legs

• Feeling very tired or weak

• Swollen gland

• A severe skin reaction (Stevens-Johnson syndrome/toxic epidermal necrolysis) like red, swollen, blistered, or peeling skin (with or without fever); red or irritated eyes; or sores in your mouth, throat, nose, or eyes

• Liver problems like dark urine, feeling tired, not hungry, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Blocadol | Dipgix | Dolac | Dolten | Dorixina forte | Duprac | Kelac | Kemanat | Keramix | Ketial | Ketodol | Ketomar | Ketorolac | Ketorolac Ahimsa | Ketorolac Fabra | Ketorolac hlb | Ketorolac labsa | Ketorolac Northia | Ketorolac Richet | Ketorolac trb pharma | Ketorolac vannier | Ketorolac vent 3 | Kpan | Sinalgico | Teledol | Tenkdol | Unicalm | Zerodol;
- (AU) Australia: Apo-ketorolac | Ketoral | Ketorolac act | Ketorolac baxter | Ketorolac juno | Ketorolac kabi | Toradol;
- (BD) Bangladesh: Actifast | Analac | Angesic | Deltora | Dolgenal | Doloket | Doltro | Emodol | Etorac | Eurolac | Kelorac | Kepros | Ket | Keteks | Ketoact | Ketoflex | Ketolab | Ketolac | Ketonaaf

| Ketoprix | Ketorin | Ketorolac | Ketoroz | Ketoshot | Kilpan | Knil | Lacor | Lopadol | Lupelac | Maxdol | Ofpain | Oradol | Orc | Pair | Perilac | Repopain | Romilac | Rotalac | Rotek | Sanoket | SB ketor | Surpim | Temoket | Todol | Tolec | Toraᑯol | Toradolin | Toralin | Toramax | Toro | Toroaid | Torosic | Winop | Xenolac | Xidolac | Xiroket | Zepac | Zeropain;

- (CH) Switzerland: Tora-dol;
- (CI) Côte d'Ivoire: Toroxim;
- (CL) Chile: Algipres | Brodifac | Burten | Dilox | Dolgenal | Ketorolaco trometamol | Netaf | Syndol;
- (CN) China: Ke duo;
- (CO) Colombia: Anaxer | Bilkinase | Keradol | Ketorolaco | Ketron | Kine;
- (CR) Costa Rica: Analgan | Brodifac | Dolgenal | Kettall;
- (DO) Dominican Republic: Analxican | Angess | Dolgenal | Dolten | Fraterolac | Iberolaco | Keradol | Ketomax | Ketorolac | Ketorolaco | Ketramol | Ketron | Supradol;
- (EC) Ecuador: Altrom | Brodifac | Burten | Dolgenal | Fadol | Kelko | Ketorolaco | Ketorolaco labovida | Ketorolaco nifa | Ketorolaco trometamina | Nomadol | Notolac | Proalgan | Rolesen | Toraᑯol;
- (EG) Egypt: Adolor | Fam | Ketolac | Ketoral | Ketrac;
- (ES) Spain: Algikey | Droal | Ketorolaco trometamol Domac | Ketorolaco trometamol Lesvi | Tonum | Toradol;
- (FI) Finland: Caloket | Toraᑯol;
- (GB) United Kingdom: Toraᑯol;
- (GT) Guatemala: Analgan | Catasil | Cidken | Dolgenal | Dolket | Kerolac | Ketorolaco | Kettall;
- (HK) Hong Kong: Ketorolac trometamol normon | Toraᑯol;
- (ID) Indonesia: Farpain | Ketorolac | Latorec | Teranol | Toraᑯol | Toramine | Trolac | Xevolac;
- (IE) Ireland: Toraᑯol;
- (IL) Israel: Topadol;
- (IN) India: Cadolac | Dolac | Fortral kt | Kelac | Kenalfin | Ket | Ketanov | Keter | Ketogon dt | Ketorol | Ketorol DT | Ketro | Ketroc | Lyxokat | Nato | Nodine | Painlok dt | T lac | Torolac | Zeroket | Zorovon;
- (IT) Italy: Lixidol | Tora Dol;
- (KE) Kenya: Dolac | Rolac;
- (KR) Korea, Republic of: Kentorac | Kerola | Kerolmin | Ketolacin | Ketoracin | Ketro | Tarasyn | Tolac | Trolac;
- (LB) Lebanon: Ketolac;
- (LT) Lithuania: Dolak | Ketanov | Ketorol | Nato;
- (LV) Latvia: Dolak | Ketanov | Ketorol | Ketorolak;
- (MX) Mexico: Ainelac | Alypharm | Apotoke | Brunacol | Celfax | Dobelor | Doket | Dolac | Dolikan | Doselmin | Efimerol | Estopein | Finlac | Gesilac | Glicima | Italker | K2sol | Katamisine | Kendol | Kendolit t | Ketanor | Ketorolaco | Ketorolaco gi kend | Ketorolaco gi kene | Ketorolaco gi serr | Ketoxil | Ketrol | Koxalgen | Lacdol | Lacdol s | Lacomin | Landaco | Lorotec | Mavidol | Mevidir | Novakerol | Onemer | Oro k | Relkedol | Rolesen | Rometran k | Supradol | Toloran | Toral | Torkol | Trodorol | Ultilap | Verolak | Vitoket | Voydol;
- (MY) Malaysia: Ketanov | Keto | Toradol;
- (NG) Nigeria: Dolac | Reals ketorolac;
- (NO) Norway: Toraᑯol;
- (PA) Panama: Dolgenal | Keradol | Supradol;
- (PE) Peru: Algias | Analgesium | Anku | Apten | Celgese | Cortalen | Dolmax | Dolnix | Dolo ket | Dolofac | Doloketal | Dolorex | Doltrex | Grabia | Halgeze | Hanalgeze | Kedol | Keradol | Ketaxal | Ketoalgin | Ketodol | Ketomax | Ketomed | Ketomolargesico | Ketopharm | Ketorolaco | Ketorolaco fmndtria | Ketorolaco MF | Ketorolaco trometamina | Ketorolin | Ketossone | Ketovet | Ketozol | Ketradol | Kine | Kitamass | Maxidol | Maxis | Q-dolese | Quetorol | Rolac | Rolesen | Sanagese | Toralac;
- (PH) Philippines: Alore | Eurolac | Ketanov | Ketero | Keto | Ketomed;
- (PK) Pakistan: K lac | Kerol | Ketin | Orkit | Toradol | Toraject | Torapan | Xevolac | Yukon;
- (PL) Poland: Toraᑯol;
- (PR) Puerto Rico: Ketorolac | Ketorolac tromethaline | Toraᑯol;
- (PT) Portugal: Toradol;

- (PY) Paraguay: Anador | Dilamol | Dilox | Dolgenal | Dolo galen | Dolomedin | Doloren | Dolostop | Dolsed | Dolten | Doxyclin | Hontocal | Kedolac | Keto odontol | Ketorolac dallas | Ketorolac empa | Ketorolac medinac | Ketorolac millet | Ketorolac pasteur | Prodent forte | Quetorol | Rapidol | Sauran forte | Tetralgin | Toralgin;
- (QA) Qatar: Fam | Ketoroget | Tora-Dol | Toradol;
- (RO) Romania: Ketanov | Ketorol;
- (RU) Russian Federation: Adolor | Dolac | Dolak | Ketalgin | Ketanov | Ketofreel | Ketokam | Ketolac | Ketorol | Ketorolac | Ketorolac Obl | Ketrodol;
- (SE) Sweden: Toradol;
- (SG) Singapore: Toradol;
- (TH) Thailand: Ketolac;
- (TR) Turkey: Ketrodol;
- (TW) Taiwan: Analif | Inco | Kelorac | Kidoton | Kop | Painoff | Sukerin | Suketon;
- (UA) Ukraine: Ambit | Blockpain | Emodol | Ketalgin | Ketalgin long | Ketanov | Ketolex | Ketolong | Ketorol | Ketorolac | Ketorolac jayson | Ketorolac norton | Ketorolac zdorovje;
- (UG) Uganda: Dolac;
- (UY) Uruguay: Dolgenal | Dolten | Eleadol;
- (VE) Venezuela, Bolivarian Republic of: Biolak | Dolak | Kelac | Ketolac | Ketorolac | Ketorolac trometamina | Ketorolaco | Kettal | Notolac;
- (VN) Viet Nam: Mildotac;
- (ZA) South Africa: Tora-dol;
- (ZM) Zambia: Dolac

REFERENCES

1. 2023 American Geriatrics Society Beers Criteria Update Expert Panel. American Geriatrics Society 2023 updated AGS Beers Criteria for potentially inappropriate medication use in older adults. *J Am Geriatr Soc*. 2023;71(7):2052-2081. doi:10.1111/jgs.18372 [PubMed 37139824]

2. Abraham NS, Hlatky MA, Antman EM, et al; ACCF/ACG/AHA. ACCF/ACG/AHA 2010 expert consensus document on the concomitant use of proton pump inhibitors and thienopyridines: a focused update of the ACCF/ACG/AHA 2008 expert consensus document on reducing the gastrointestinal risks of antiplatelet therapy and NSAID use: a report of the American College of Cardiology Foundation Task Force on Expert Consensus Documents. *Circulation.* 2010;122(24):2619-2633. doi:10.1161/CIR.0b013e318202f70 [PubMed 21060077]

3. Adler GK, Stowasser M, Correa RR, et al. Primary aldosteronism: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2025;110(9):2453-2495. doi:10.1210/clinem/dgaf284 [PubMed 40658480]

4. Alam I, Ferrell LD, Bass NM. Vanishing bile duct syndrome temporally associated with ibuprofen use. *Am J Gastroenterol*. 1996;91(8):1626-1630. [PubMed 8759674]

5. Aldrink JH, Ma M, Wang W, Caniano DA, Wispe J, Puthoff T. Safety of ketorolac in surgical neonates and infants 0 to 3 months old. *J Pediatr Surg*. 2011;46(6):1081-1085

6. Alqahtani Z, Jamali F. Clinical Outcomes of aspirin interaction with other non-steroidal anti-inflammatory drugs: A systematic review. *J Pharm Pharm Sci*. 2018;21(1s):29854. doi:10.18433/jpps29854 [PubMed 29891025]

7. Alsabri MAH, Diab RA, Alqeeq BF, et al. Ketorolac as an analgesic in pediatric acute pain management: a systematic review and meta-analysis. *Pediatr Emerg Care*. 2025;41(7):e34-e47. doi:10.1097/PEC.0000000000003381 [PubMed 40326658]

8. American Academy of Pediatrics Committee on Fetus and Newborn; American Academy of Pediatrics Section on Surgery; Canadian Paediatric Society Fetus and Newborn Committee, et al, ”Prevention and Management of Pain in the Neonate: An Update,” *Pediatrics*, 2006, 118(5):2231-41. [PubMed 17079598]

9. American College of Obstetricians and Gynecologists (ACOG). ACOG Practice Bulletin No. 209: Obstetric analgesia and anesthesia. *Obstet Gynecol*. 2019;133(3):e208-e225. [PubMed 30801474]

10. American College of Surgeons (ACS). ACS Trauma Quality Programs best practices guidelines for acute pain management in trauma patients. https://www.facs.org/-/media/files/quality-programs/trauma/tqip/acute_pain_guidelines.ashx. Published November 2020. Accessed August 30, 2021.

11. Amsterdam EA, Wenger NK, Brindis RG, et al. 2014 AHA/ACC Guideline for the Management of Patients with Non-ST-Elevation Acute Coronary Syndromes: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2014;64(24):e139-e228. doi:10.1016/j.jacc.2014.09.017. Erratum in: *J Am Coll Cardiol*. 2014;64(24):2713-2714. Dosage error in article text. [PubMed 25260718]

12. Anderson PO, Sauberan JB. Modeling drug passage into human milk. *Clin Pharmacol Ther*. 2016;100(1):42-52. [PubMed 27060684]

13. Andrejak M, Davion T, Gineston JL, Capron JP. Cross hepatotoxicity between non-steroidal anti-inflammatory drugs. *Br Med J (Clin Res Ed)*. 1987;295(6591):180-181. doi:10.1136/bmj.295.6591.180 [PubMed 3115366]

14. Anglin R, Yuan Y, Moayyedi P, Tse F, Armstrong D, Leontiadis GI. Risk of upper gastrointestinal bleeding with selective serotonin reuptake inhibitors with or without concurrent nonsteroidal anti-inflammatory use: a systematic review and meta-analysis. *Am J Gastroenterol*. 2014;109(6):811-819. doi:10.1038/ajg.2014.82 [PubMed 24777151]

15. Aronoff GR, Bennett WM, Berns JS, et al. *Drug Prescribing in Renal Failure: Dosing Guidelines for Adults and Children, 5th ed*. Philadelphia, PA: American College of Physicians; 2007.

16. Baker M, Perazella MA. NSAIDs in CKD: are they safe? *Am J Kidney Dis*. 2020;76(4):546-557. doi:10.1053/j.ajkd.2020.03.023 [PubMed 32479922]

17. Bally M, Dendukuri N, Rich B, et al. Risk of acute myocardial infarction with NSAIDs in real world use: bayesian meta-analysis of individual patient data. *BMJ*. 2017;357:j1909. doi:10.1136/bmj.j1909 [PubMed 28487435]

18. Barbaryan A, Iyinagoro C, Nwankwo N, et al. Ibuprofen-induced hemolytic anemia. *Case Rep Hematol*. 2013;2013:142865. doi:10.1155/2013/142865 [PubMed 23710383]

19. Battistella M, Mamdami MM, Juurlink DN, Rabeneck L, Laupacis A. Risk of upper gastrointestinal hemorrhage in warfarin users treated with nonselective NSAIDs or COX-2 inhibitors. *Arch Intern Med*. 2005;165(2):189-192. doi:10.1001/archinte.165.2.189 [PubMed 15668365]

20. Bermas BL. Non-steroidal anti inflammatory drugs, glucocorticoids and disease modifying anti-rheumatic drugs for the management of rheumatoid arthritis before and during pregnancy. *Curr Opin Rheumatol*. 2014;26(3):334-340. [PubMed 24663106]

21. Bhala N, Emberson J, Merhi A, et al; Coxib and traditional NSAID Trialists' (CNT) Collaboration. Vascular and upper gastrointestinal effects of non-steroidal anti-inflammatory drugs: meta-analyses of individual participant data from randomised trials. *Lancet*. 2013;382(9894):769-779. doi:10.1016/S0140-6736(13)60900-9 [PubMed 23726390]

22. Bhangu A, Singh P, Fitzgerald JE, Slesser A, Tekkis P. Postoperative nonsteroidal anti-inflammatory drugs and risk of anastomotic leak: meta-analysis of clinical and experimental studies. *World J Surg*. 2014;38(9):2247-2257. doi:10.1007/s00268-014-2531-1 [PubMed 24682313]

23. Bhatt DL, Scheiman J, Abraham NS, et al; American College of Cardiology Foundation Task Force on Clinical Expert Consensus Documents. ACCF/ACG/AHA 2008 expert consensus document on reducing the gastrointestinal risks of antiplatelet therapy and NSAID use: a report of the American College of Cardiology Foundation Task Force on Clinical Expert Consensus

Documents. *Circulation*. 2008;118(18):1894-1909. doi:10.1161/CIRCULATIONAHA.108.191087 [PubMed 18836135]

24. Biggins SW, Angeli P, Garcia-Tsao G, et al. Diagnosis, evaluation, and management of ascites, spontaneous bacterial peritonitis and hepatorenal syndrome: 2021 practice guidance by the American Association for the Study of Liver Diseases. *Hepatology*. 2021;74(2):1014-1048. doi:10.1002/hep.31884 [PubMed 33942342]

25. Bloor M, Paech M. Nonsteroidal anti-inflammatory drugs during pregnancy and the initiation of lactation. *Anesth Analg*. 2013;116(5):1063-1075. [PubMed 23558845]

26. Brockow K, Przybilla B, Aberer W, et al. Guideline for the diagnosis of drug hypersensitivity reactions: S2K-Guideline of the German Society for Allergology and Clinical Immunology (DGAKI) and the German Dermatological Society (DDG) in collaboration with the Association of German Allergologists (AeDA), the German Society for Pediatric Allergology and Environmental Medicine (GPA), the German Contact Dermatitis Research Group (DKG), the Swiss Society for Allergy and Immunology (SGAI), the Austrian Society for Allergology and Immunology (ÖGAI), the German Academy of Allergology and Environmental Medicine (DAAU), the German Center for Documentation of Severe Skin Reactions and the German Federal Institute for Drugs and Medical Products (BfArM). *Allergo J Int*. 2015;24(3):94-105. doi:10.1007/s40629-015-0052-6 [PubMed 26120552]

27. Brophy PD. Changing the paradigm in pediatric acute kidney injury. *J Pediatr*. 2013;162(6):1094-1096. doi:10.1016/j.jpeds.2013.01.065 [PubMed 23453549]

28. Buck ML. Clinical Experience With Ketorolac in Children. *Ann Pharmacother*. 1994;28(9):1009-1013. [PubMed 7803871]

29. Buckley MM and Brogden RN, ”Ketorolac. A Review of Its Pharmacodynamic and Pharmacokinetic Properties, and Therapeutic Potential,” *Drugs*, 1990, 39(1):86-109. [PubMed 2178916]

30. Burd RS and Tobias JD, ”Ketorolac for Pain Management After Abdominal Surgical Procedures in Infants,” *South Med J*, 2002, 95(3):331-3. [PubMed 11902701]

31. Capone ML, Sciulli MG, Tacconelli S, et al, “Pharmacodynamic Interaction of Naproxen With Low-Dose Aspirin in Healthy Subjects,” *J Am Coll Cardiol*, 2005, 45(8):1295-301. [PubMed 15837265]

32. Capone ML, Tacconelli S, Sciulli MG, et al. Clinical pharmacology of platelet, monocyte, and vascular cyclooxygenase inhibition by naproxen and low-dose aspirin in healthy subjects. *Circulation*. 2004;109(12):1468-1471. doi:10.1161/01.CIR.0000124715.27937.78 [PubMed 15037526]

33. Carpuject (ketorolac tromethamine) [prescribing information]. Lake Forest, IL: Hospira, Inc; March 2021.

34. Castellsague J, Riera-Guardia N, Calingaert B, et al; Safety of Non-Steroidal Anti-Inflammatory Drugs (SOS) Project. Individual NSAIDs and upper gastrointestinal complications: a systematic review and meta-analysis of observational studies (the SOS project). *Drug Saf*. 2012;35(12):1127-1146. doi:10.2165/11633470-000000000-00000 [PubMed 23137151]

35. Caughey GE, Cleland LG, Penglis PS, Gamble JR, James MJ. Roles of cyclooxygenase (COX)-1 and COX-2 in prostanoid production by human endothelial cells: selective up-regulation of prostacyclin synthesis by COX-2. *J Immunol*. 2001;167(5):2831-2838. doi:10.4049/jimmunol.167.5.2831 [PubMed 11509629]

36. Cawthorn TR, Phelan R, Davidson JS, Turner KE. Retrospective analysis of perioperative ketorolac and postoperative bleeding in reduction mammoplasty. *Can J Anaesth*. 2012;59(5):466-472. doi:10.1007/s12630-012-9682-z [PubMed 22434401]

37. Chalasani NP, Hayashi PH, Bonkovsky HL, et al. ACG Clinical Guideline: the diagnosis and management of idiosyncratic drug-induced liver injury. *Am J Gastroenterol*. 2014;109(7):950-966. [PubMed 24935270]

38. Chan AT, Manson JE, Albert CM, et al. Nonsteroidal antiinflammatory drugs, acetaminophen, and the risk of cardiovascular events. *Circulation*. 2006;113(12):1578-1587. doi:10.1161/CIRCULATIONAHA.105 [PubMed 16534006]

39. Chan DK, Parikh SR. Perioperative ketorolac increases post-tonsillectomy hemorrhage in adults but not children. *Laryngoscope*. 2014;124(8):1789-1793. doi:10.1002/lary.24555 [PubMed 24338331]

40. Chan J, Bajnath A, Fromkin B, et al. Ketorolac prescribing practices in an acute care hospital and the incidence of acute renal failure. *World Journal of Nephrology & Urology*. 2014;3(3):113-117.

41. Chandok N, Watt KD. Pain management in the cirrhotic patient: the clinical challenge. *Mayo Clin Proc.* 2010;85(5):451-458. doi:10.4065/mcp.2009.0534 [PubMed 20357277]

42. Chen AH, Bennett CR. Ketorolac-induced bronchospasm in an aspirin-intolerant patient. *Anesth Prog*. 1994;41(4):102-107. [PubMed 8934961]

43. Cheng Y, Austin SC, Rocca B, et al. Role of prostacyclin in the cardiovascular response to thromboxane A2. *Science*. 2002;296(5567):539-541. doi:10.1126/science.1068711 [PubMed 11964481]

44. Chou CI, Shih CJ, Chen YT, et al. Adverse effects of oral nonselective and cyclooxygenase-2-selective NSAIDs on hospitalization for acute kidney injury: A nested case-control cohort study. *Medicine (Baltimore)*. 2016;95(9):e2645. doi:10.1097/MD.0000000000002645 [PubMed 26945352]

45. Chou R, Gordon DB, de Leon-Casasola OA, et al. Management of postoperative pain: a clinical practice guideline from the American Pain Society, the American Society of Regional Anesthesia and Pain Medicine, and the American Society of Anesthesiologists' Committee on Regional Anesthesia, Executive Committee, and Administrative Council [published correction appears in *J Pain*. 2016;17(4):508-510]. *J Pain*. 2016;17(2):131-157. doi: 10.1016/j.jpain.2015.12.008. [PubMed 26827847]

46. Conrad KA, Fagan TC, Mackie MJ, Mayshar PV. Effects of ketorolac tromethamine on hemostasis in volunteers. *Clin Pharmacol Ther*. 1988;43(5):542-546. doi:10.1038/clpt.1988.70 [PubMed 3259170]

47. Corelli RL, Gericke KR. Renal insufficiency associated with intramuscular administration of ketorolac tromethamine. *Ann Pharmacother*. 1993;27(9):1055-1057. doi:10.1177/106002809302700908 [PubMed 8219436]

48. Cryer B, Feldman M. Cyclooxygenase-1 and cyclooxygenase-2 selectivity of widely used nonsteroidal anti-inflammatory drugs. *Am J Med*. 1998;104(5):413-421. doi:10.1016/s0002-9343(98)00091-6 [PubMed 9626023]

49. Dathe K, Hultzsch S, Pritchard LW, Schaefer C. Risk estimation of fetal adverse effects after short-term second trimester exposure to non-steroidal anti-inflammatory drugs: a literature review. *Eur J Clin Pharmacol*. 2019;75(10):1347-1353. doi:10.1007/s00228-019-02712-2 [PubMed 31273431]

50. Davidson BL, Verheijen S, Lensing AW, et al. Bleeding risk of patients with acute venous thromboembolism taking nonsteroidal anti-inflammatory drugs or aspirin. *JAMA Intern Med*. 2014;174(6):947-953. doi:10.1001/jamainternmed.2014.946 [PubMed 24733305]

51. Dawkins TN, Barclay CA, Gardiner RL, et al, "Safety of Intravenous Use of Ketorolac in Infants Following Cardiothoracic Surgery," *Cardiol Young*, 2009, 19(1):105-8. [PubMed 19134246]

52. Delzer LM, Golightly LK, Kiser TH, Biggins SW, Lewis VJ, Kim II. Calcineurin inhibitor and nonsteroidal anti-inflammatory drug interaction: Implications of changes in renal function associated with concurrent use. *J Clin Pharmacol*. 2018;58(11):1443-1451. doi:10.1002/jcph.1264 [PubMed 29799625]

53. Doña I, Blanca-López N, Jagemann LR, et al. Response to a selective COX-2 inhibitor in patients with urticaria/angioedema induced by nonsteroidal anti-inflammatory drugs. *Allergy*. 2011;66(11):1428-1433. doi:10.1111/j.1398-9995.2011.02684.x [PubMed 21834936]

54. Donati M, Conforti A, Lenti MC, et al; DILI-IT Study Group. Risk of acute and serious liver injury associated to nimesulide and other NSAIDs: data from drug-induced liver injury case-control study in Italy. *Br J Clin Pharmacol*. 2016;82(1):238-248. doi:10.1111/bcp.12938 [PubMed 26991794]

55. Dsida RM, Wheeler M, Birmingham PK, et al, "Age-Stratified Pharmacokinetics of Ketorolac Tromethamine in Pediatric Surgical Patients," *Anesth Analg*, 2002, 94(2):266-70. [PubMed 11812682]

56. Dwyer JP, Jayasekera C, Nicoll A. Analgesia for the cirrhotic patient: a literature review and recommendations. *J Gastroenterol Hepatol*. 2014;29(7):1356-1360. doi:10.1111/jgh.12560 [PubMed 24548074]

57. Engoren MC, Habib RH, Zacharias A, et al. Postoperative analgesia with ketorolac is associated with decreased mortality after isolated coronary artery bypass graft surgery in patients already receiving aspirin: a propensity-matched study. *J Cardiothorac Vasc Anesth*. 2007;21(6):820-826. doi:10.1053/j.jvca.2007.01.024 [PubMed 18068059]

58. Esteves A, Teixeira da Silva F, Carvalho J, Carvoeiro A, Felgueiras P. Diclofenac-induced immune hemolytic anemia: A case report and review of literature. *Cureus*. 2021;13(1):e12903. doi:10.7759/cureus.12903 [PubMed 33654588]

59. Etches RC, Warriner CB, Badner N, et al. Continuous intravenous administration of ketorolac reduces pain and morphine consumption after total hip or knee arthroplasty. *Anesth Analg*. 1995;81(6):1175-1180. [PubMed 7486100]

60. Expert opinion. Senior Hepatic Editorial Team: Matt Harris, PharmD, MHS, BCPS, FAST, FCCP; Jeong Park, PharmD, MS, BCTXP, FCCP, FAST; Arun Jesudian, MD; Sasan Sakiani, MD.

61. Expert opinion. Senior Renal Editorial Team: Bruce Mueller, PharmD, FCCP, FASN, FNKF; Jason A. Roberts, PhD, BPharm (Hons), B App Sc, FSHP, FISAC; Michael Heung, MD, MS.

62. Famularo G, Bizzarri C, Nicotra GC. Acute pancreatitis caused by ketorolac tromethamine. *J Clin Gastroenterol*. 2002;34(3):283-284. doi:10.1097/00004836-200203000-00021 [PubMed 11873116]

63. Fernandes V, Alfaro TM, Baptista JP, Regateiro FS, Fradinho F, Cordeiro CR. Severe ketorolac-induced asthma diagnosed by chest computed tomography. *J Thorac Dis*. 2017;9(Suppl 16):S1567-S1569. doi:10.21037/jtd.2017.11.36 [PubMed 29255644]

64. Flamm SL, Wong F, Ahn J, Kamath PS. AGA clinical practice update on the evaluation and management of acute kidney injury in patients with cirrhosis: expert review. *Clin Gastroenterol Hepatol.* 2022;20(12):2707-2716. doi:10.1016/j.cgh.2022.08.033 [PubMed 36075500]

65. Fong J, Gora ML. Reversible renal insufficiency following ketorolac therapy. *Ann Pharmacother*. 1993;27(4):510-512. doi:10.1177/106002809302700422 [PubMed 8477132]

66. Forrest JB, Heitlinger EL, Revell S. Ketorolac for postoperative pain management in children. *Drug Saf*. 1997;16(5):309-329. [PubMed 9187531]

67. Franz CC, Hildbrand C, Born C, Egger S, Rätz Bravo AE, Krähenbühl S. Dose adjustment in patients with liver cirrhosis: impact on adverse drug reactions and hospitalizations. *Eur J Clin Pharmacol.* 2013;69(8):1565-1573. doi:10.1007/s00228-013-1502-z [PubMed 23588561]

68. Friedman BW, Garber L, Yoon A, et al. Randomized trial of IV valproate vs metoclopramide vs ketorolac for acute migraine. *Neurology*. 2014;82(11):976-983. doi:10.1212/WNL.0000000000000223 [PubMed 24523483]

69. Friedrichsdorf SJ, Goubert L. Pediatric pain treatment and prevention for hospitalized children. *Pain Rep*. 2019;5(1):e804. doi:10.1097/PR9.0000000000000804 [PubMed 32072099]

70. Fuller DK, Kalekas PJ. Ketorolac and gastrointestinal ulceration. *Ann Pharmacother*. 1993;27(7-8):978-979. doi:10.1177/106002809302700731 [PubMed 8364288]

71. García Rodríguez LA, Cattaruzzi C, Troncon MG, Agostinis L. Risk of hospitalization for upper gastrointestinal tract bleeding associated with ketorolac, other nonsteroidal anti-inflammatory drugs, calcium antagonists, and other antihypertensive drugs. *Arch Intern Med*. 1998;158(1):33-39. doi:10.1001/archinte.158.1.33 [PubMed 9437376]

72. Garratty G, Arndt PA. Drugs that have been shown to cause drug-induced immune hemolytic anemia or positive direct antiglobulin tests: some interesting findings since 2007. *Immunohematology*. 2014;30(2):66-79. [PubMed 25247621]

73. Gladding PA, Webster MW, Farrell HB, Zeng IS, Park R, Ruijne N. The antiplatelet effect of six non-steroidal anti-inflammatory drugs and their pharmacodynamic interaction with aspirin in healthy volunteers. *Am J Cardiol*. 2008;101(7):1060-1063. doi:10.1016/j.amjcard.2007.11.054 [PubMed 18359332]

74. Gobble RM, Hoang HLT, Kachniarz B, Orgill DP. Ketorolac does not increase perioperative bleeding: a meta-analysis of randomized controlled trials. *Plast Reconstr Surg*. 2014;133(3):741-755. doi:10.1097/01.prs.0000438459.60474.b5 [PubMed 24572864]

75. Goetz CM, Sterchele JA, Harchelroad FP. Anaphylactoid reaction following ketorolac tromethamine administration. *Ann Pharmacother*. 1992;26(10):1237-1238. doi:10.1177/106002809202601008 [PubMed 1421646]

76. Golightly LK, Teitelbaum I, Kiser TH, et al, eds. *Renal Pharmacotherapy*. New York, NY: Springer Science; 2013

77. Greer IA. Effects of ketorolac tromethamine on hemostasis. *Pharmacotherapy*. 1990;10(6 (Pt 2)):71S-76S. [PubMed 2082316]

78. Gulmez SE, Larrey D, Pageaux GP, et al. Transplantation for acute liver failure in patients exposed to NSAIDs or paracetamol (acetaminophen): the multinational case-population SALT study. *Drug Saf*. 2013;36(2):135-144. doi:10.1007/s40264-012-0013-7 [PubMed 23325533]

79. Gulmez SE, Unal US, Lassalle R, Chartier A, Grolleau A, Moore N. Risk of hospital admission for liver injury in users of NSAIDs and nonoverdose paracetamol: Preliminary results from the EPIHAM study. *Pharmacoepidemiol Drug Saf*. 2018;27(11):1174-1181. doi:10.1002/pds.4640 [PubMed 30112779]

80. Gupta A, Daggett C, Drant S, et al, ”Prospective Randomized Trial of Ketorolac After Congenital Heart Surgery,” *J Cardiothorac Vasc Anesth*, 2004, 18(4):454-7. [PubMed 15365927]

81. Gupta A, Daggett C, Ludwick J, et al, ”Ketorolac After Congenital Heart Surgery: Does It Increase the Risk of Significant Bleeding Complications?” *Paediatr Anaesth*, 2005, 15(2):139-42. [PubMed 15675931]

82. Gupta AK, Parker BM. Bleeding ater a single dose of ketorolac in a postoperative patient. *Cureus*. 2020;12(6):e8919. doi:10.7759/cureus.8919 [PubMed 32760620]

83. Hadley EE, Monsivais L, Pacheco L, et al. Multimodal pain management for cesarean delivery: a double-blinded, placebo-controlled, randomized clinical trial. *Am J Perinatol*. 2019;36(11):1097-1105. doi:10.1055/s-0039-1681096 [PubMed 30822800]

84. Hall ST, Mangram AJ, Barletta JF. Identification of risk factors for acute kidney injury from intravenous ketorolac in geriatric trauma patients. *World J Surg*. 2022;46(1):98-103. doi:10.1007/s00268-021-06320-z [PubMed 34553259]

85. Haragsim L, Dalal R, Bagga H, Bastani B. Ketorolac-induced acute renal failure and hyperkalemia: report of three cases. *Am J Kidney Dis*. 1994;24(4):578-580. doi:10.1016/s0272-6386(12)80215-0 [PubMed 7942813]

86. Hebert WG, Scopelitis E. Ketorolac-precipitated asthma. *South Med J*. 1994;87(2):282-283. doi:10.1097/00007611-199402000-00030 [PubMed 8115903]

87. Heidenreich PA, Bozkurt B, Aguilar D, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *Circulation*. 2022;145(18):e895-e1032. doi:10.1161/CIR.0000000000001063. Erratum in: *Circulation*. 2022;145(18):e1033. Erratum in: *Circulation*. 2022;146(13):e185. Erratum in: *Circulation*. 2023;147(14):e674. [PubMed 35363499]

88. Hennessy S, Kinman JL, Berlin JA, et al. Lack of hepatotoxic effects of parenteral ketorolac in the hospital setting. *Arch Intern Med*. 1997;157(21):2510-2514. [PubMed 9385304]

89. Hilscher MB, Odell LJ, Myhre LJ, Prokop L, Talwalkar J. The pharmacotherapy of cirrhosis: concerns and proposed investigations and solutions. *J Clin Pharm Ther.* 2016;41(6):587-591. doi:10.1111/jcpt.12443 [PubMed 27576303]

90. Hong Y, Gengo FM, Rainka MM, Bates VE, Mager DE. Population pharmacodynamic modelling of aspirin- and Ibuprofen-induced inhibition of platelet aggregation in healthy subjects. *Clin Pharmacokinet*. 2008;47(2):129-137. doi:10.2165/00003088-200847020-00006 [PubMed 18193919]

91. Horsley RD, Vogels ED, McField DAP, et al. Multimodal postoperative pain control is effective and reduces opioid use after laparoscopic Roux-en-Y gastric bypass. *Obes Surg*. 2019;29(2):394-400. doi: 10.1007/s11695-018-3526-z. [PubMed 30317488]

92. Hostage J, Kolettis D, Sverdlov D, et al. Increased scheduled intravenous ketorolac after cesarean delivery and its effect on opioid use: a randomized controlled trial. *Obstet Gynecol*. 2023;141(4):783-790. doi:10.1097/AOG.0000000000005120 [PubMed 36897140]

93. Howard ML, Isaacs AN, Nisly SA. Continuous infusion nonsteroidal anti-inflammatory drugs for perioperative pain management. *J Pharm Pract*. 2018;31(1):66-81. [PubMed 27580638]

94. Howard ML, Warhurst RD, Sheehan C. Safety of continuous infusion ketorolac in postoperative coronary artery bypass graft surgery patients. *Pharmacy (Basel)*. 2016;4(3):22. doi:10.3390/pharmacy4030022 [PubMed 28970395]

95. Huang SP, Wen YC, Huang ST, Lin CW, Wang TD, Hsiao FY. Nonsteroidal anti-inflammatory drugs and risk of first hospitalization for heart failure in patients with no history of heart failure: A population-based case-crossover study. *Drug Saf*. 2019;42(1):67-75. doi:10.1007/s40264-018-0720-9 [PubMed 30232741]

96. Huerta C, Castellsague J, Varas-Lorenzo C, García Rodríguez LA. Nonsteroidal anti-inflammatory drugs and risk of ARF in the general population. *Am J Kidney Dis*. 2005;45(3):531-539. doi:10.1053/j.ajkd.2004.12.005 [PubMed 15754275]

97. Ito S. Drug therapy for breast-feeding women. *N Engl J Med*. 2000;343(2):118-126. [PubMed 10891521]

98. Javaherforooshzadeh F, Abdalbeygi H, Janatmakan F, Gholizadeh B. Comparing the effects of ketorolac and *Paracetamol* on postoperative pain relief after coronary artery bypass graft surgery. A randomized clinical trial. *J Cardiothorac Surg*. 2020;15(1):80. doi:10.1186/s13019-020-01125-y [PubMed 32393370]

99. Javier Rodríguez-González F, Montero JL, Puente J, et al. Orthotopic liver transplantation after subacute liver failure induced by therapeutic doses of ibuprofen. *Am J Gastroenterol*. 2002;97(9):2476-2477. doi:10.1111/j.1572-0241.2002.06015.x [PubMed 12358284]

100. Judkins JH, Dray TG, Hubbell RN. Intraoperative ketorolac and posttonsillectomy bleeding. *Arch Otolaryngol Head Neck Surg*. 1996;122(9):937-940. doi:10.1001/archotol.1996.01890210017004 [PubMed 8797556]

101. Kean WF, Lock CJ, Rischke J, Butt R, Buchanan WW, Howard-Lock H. Effect of R and S enantiomers of naproxen on aggregation and thromboxane production in human platelets. *J Pharm Sci*. 1989;78(4):324-327. doi:10.1002/jps.2600780413 [PubMed 2724096]

102. Ketorolac tromethamine injection vial, USP [prescribing information]. Deerfield, IL: Baxter Healthcare Corporation; September 2024.

103. Ketorolac tromethamine injection, USP [product monograph]. Toronto, Ontario, Canada: Fresenius Kabi Canada Ltd; December 2022.

104. Ketorolac tromethamine injection prefilled syringe [prescribing information]. Lake Zurich, IL: Fresenius Kabi; May 2022.

105. Ketorolac tromethamine tablets [prescribing information]. Pennington, NJ: Zydus Pharmaceuticals (USA) Inc; July 2024.

106. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int*. 2013;3(suppl):1-150. https://kdigo.org/wp-content/uploads/2017/02/KDIGO_2012_CKD_GL.pdf.

107. Kim DH, Jeon YT, Kim HG, et al. Comparison between ketorolac- and fentanyl-based patient-controlled analgesia for acute kidney injury after robot-assisted radical prostatectomy: a retrospective propensity score-matched analysis. *World J Urol*. 2023;41(5):1437-1444. doi:10.1007/s00345-023-04374-3 [PubMed 37004573]

108. Kim YJ, Lim KH, Kim MY, et al. Cross-reactivity to acetaminophen and celecoxib according to the type of nonsteroidal anti-inflammatory drug hypersensitivity. *Allergy Asthma Immunol Res*. 2014;6(2):156-162. doi:10.4168/aair.2014.6.2.156 [PubMed 24587953]

109. Klein M, Støckel M, Rosenberg J, Gögenur I. Intraoperative ketorolac and bleeding after laparoscopic Roux-en-Y gastric by-pass surgery. *Acta Chir Belg*. 2012;112(5):369-373. doi:10.1080/00015458.2012.11680854 [PubMed 23175926]

110. Knowles SR, Drucker AM, Weber EA, Shear NH. Management options for patients with aspirin and nonsteroidal antiinflammatory drug sensitivity. *Ann Pharmacother*. 2007;41(7):1191-1200. doi:10.1345/aph.1K023 [PubMed 17609236]

111. Kohli P, Steg PG, Cannon CP, et al; REACH Registry Investigators. NSAID use and association with cardiovascular outcomes in outpatients with stable atherothrombotic disease. *Am J Med*. 2014;127(1):53-60.e1. doi:10.1016/j.amjmed.2013.08.017 [PubMed 24280110]

112. Kowalski ML, Agache I, Bavbek S, et al. Diagnosis and management of NSAID-Exacerbated Respiratory Disease (N-ERD)-a EAACI position paper. *Allergy*. 2019;74(1):28-39. doi:10.1111/all.13599 [PubMed 30216468]

113. Kowalski ML, Asero R, Bavbek S, et al. Classification and practical approach to the diagnosis and management of hypersensitivity to nonsteroidal anti-inflammatory drugs. *Allergy*. 2013;68(10):1219-1232. doi:10.1111/all.12260 [PubMed 24117484]

114. Kowalski ML, Makowska JS. Seven steps to the diagnosis of NSAIDs hypersensitivity: how to apply a new classification in real practice? *Allergy Asthma Immunol Res*. 2015;7(4):312-320. doi:10.4168/aair.2015.7.4.312 [PubMed 25749768]

115. Kowalski ML, Makowska JS, Blanca M, et al. Hypersensitivity to nonsteroidal anti-inflammatory drugs (NSAIDs) - classification, diagnosis and management: review of the EAACI/ENDA(#) and GA2LEN/HANNA*. *Allergy*. 2011;66(7):818-829. doi:10.1111/j.1398-9995.2011.02557.x [PubMed 21631520]

116. Kraemer FW, Rose JB. Pharmacologic management of acute pediatric pain. *Anesthesiol Clin*. 2009;27(2):241-268. doi:10.1016/j.anclin.2009.07.002 [PubMed 19703675]

117. Kulo A, Smits A, Maleškić S, Van de Velde M, Van Calsteren K, De Hoon J, Verbesselt R, Deprest J, Allegaert K. Enantiomer-specific ketorolac pharmacokinetics in young women, including pregnancy and postpartum period. *Bosn J Basic Med Sci*. 2017;17(1):54-60. doi:10.17305/bjbms.2016.1515 [PubMed 27968707]

118. Kurella M, Bennett WM, Chertow GM. Analgesia in patients with ESRD: a review of available evidence. *Am J Kidney Dis*. 2003;42(2):217-228. doi:10.1016/s0272-6386(03)00645-0 [PubMed

12900801]

119. Laidlaw TM, Cahill KN. Current knowledge and management of hypersensitivity to aspirin and NSAIDs. *J Allergy Clin Immunol Pract*. 2017;5(3):537-545. doi:10.1016/j.jaip.2016.10.021 [PubMed 28483309]

120. Laine L, Curtis SP, Cryer B, Kaur A, Cannon CP. Risk factors for NSAID-associated upper GI clinical events in a long-term prospective study of 34 701 arthritis patients. *Aliment Pharmacol Ther*. 2010;32(10):1240-1248. doi:10.1111/j.1365-2036.2010.04465.x [PubMed 20955443]

121. Lamberts M, Lip GY, Hansen ML, et al. Relation of nonsteroidal anti-inflammatory drugs to serious bleeding and thromboembolism risk in patients with atrial fibrillation receiving antithrombotic therapy: a nationwide cohort study. *Ann Intern Med*. 2014;161(10):690-698. doi:10.7326/M13-1581 [PubMed 25402512]

122. Lapi F, Azoulay L, Yin H, Nessim SJ, Suissa S. Concurrent use of diuretics, angiotensin converting enzyme inhibitors, and angiotensin receptor blockers with non-steroidal anti-inflammatory drugs and risk of acute kidney injury: nested case-control study. *BMJ*. 2013;346:e8525. doi:10.1136/bmj.e8525 [PubMed 23299844]

123. Lee J, Zhou EP, Davis RL, Ouyang Y, Lin HM, Yudkowitz FS. Bleeding and ketorolac use in pediatric circumcision. *Paediatr Anaesth*. 2023;33(6):481-485. doi:10.1111/pan.14661 [PubMed 36892424]

124. Lee YC, Chang CH, Lin JW, Chen HC, Lin MS, Lai MS. Non-steroidal anti-inflammatory drugs use and risk of upper gastrointestinal adverse events in cirrhotic patients. *Liver Int.* 2012;32(5):859-866. doi:10.1111/j.1478-3231.2011.02739.x [PubMed 22226322]

125. Lewis JH, Stine JG. Review article: prescribing medications in patients with cirrhosis - a practical guide. *Aliment Pharmacol Ther.* 2013;37(12):1132-1156. doi:10.1111/apt.12324 [PubMed 23638982]

126. Lynn AM, Bradford H, Kantor ED, et al, ”Postoperative Ketorolac Tromethamine Use in Infants Aged 6-18 Months: The Effect on Morphine Usage, Safety Assessment, and Stereo-Specific Pharmacokinetics,” *Anesth Analg*, 2007, 104(5):1040-51. [PubMed 17456651]

127. Maliekal J, Elboim CM. Gastrointestinal complications associated with intramuscular ketorolac tromethamine therapy in the elderly. *Ann Pharmacother*. 1995;29(7-8):698-701. doi:10.1177/10600280950 808 [PubMed 8520083]

128. Mariano F, Cogno C, Giaretta F, et al. Urinary protein profiles in ketorolac-associated acute kidney injury in patients undergoing orthopedic day surgery. *Int J Nephrol Renovasc Dis*. 2017;10:269-274. doi:10.2147/IJNRD.S137102 [PubMed 29075132]

129. Martin E, Vickers B, Landau R, Reece-Stremtan S. ABM clinical protocol #28, peripartum analgesia and anesthesia for the breastfeeding mother. *Breastfeed Med*. 2018;13(3):164-171. doi:10.1089/bfm.2018.29087.ejm [PubMed 29595994]

130. McLay JS, Engelhardt T, Mohammed BS, et al. The pharmacokinetics of intravenous ketorolac in children aged 2 months to 16 years: A population analysis. *Paediatr Anaesth*. 2018;28(2):80-86. doi:10.1111/pan.13302 [PubMed 29266539]

131. Mechanick JI, Apovian C, Brethauer S, et al. Clinical practice guidelines for the perioperative nutrition, metabolic, and nonsurgical support of patients undergoing bariatric procedures - 2019 update: cosponsored by American Association of Clinical Endocrinologists/American College of Endocrinology, the Obesity Society, American Society for Metabolic & Bariatric Surgery, Obesity Medicine Association, and American Society of Anesthesiologists. *Surg Obes Relat Dis*. 2020;16(2):175-247. doi:10.1016/j.soard.2019.10.025 [PubMed 31917200]

132. Menniti-Ippolito F, Maggini M, Raschetti R, Da Cas R, Traversa G, Walker AM. Ketorolac use in outpatients and gastrointestinal hospitalization: a comparison with other non-steroidal anti-inflammatory drugs in Italy. *Eur J Clin Pharmacol*. 1998;54(5):393-397. doi:10.1007/s002280050481 [PubMed 9754982]

133. Micu MC, Micu R, Ostensen M. Luteinized unruptured follicle syndrome increased by inactive disease and selective cyclooxygenase 2 inhibitors in women with inflammatory arthropathies. *Arthritis Care Res (Hoboken)*. 2011;63(9):1334-1338. doi:10.1002/acr.20510 [PubMed 21618455]

134. Miescher PA. Blood dyscrasias secondary to non-steroidal anti-inflammatory drugs. *Med Toxicol*. 1986;1 Suppl 1:57-70. [PubMed 3547002]

135. Minaldi E, Cahill K. Recent updates in understanding NSAID hypersensitivity. *Curr Allergy Asthma Rep*. 2023;23(3):181-188. doi:10.1007/s11882-023-01064-3 [PubMed 36757490]

136. Misurac JM, Knoderer CA, Leiser JD, Nailescu C, Wilson AC, Andreoli SP. Nonsteroidal anti-inflammatory drugs are an important cause of acute kidney injury in children. *J Pediatr*. 2013;162(6):1153-1159, 1159.e1. doi:10.1016/j.jpeds.2012.11.069 [PubMed 23360563]

137. Moffett BS, Cabrera A. Ketorolac-associated renal morbidity: risk factors in cardiac surgical infants. *Cardiol Young*. 2013;23(5):752-754. [PubMed 23088994]

138. Moffett BS, Wann TI, Carberry KE, et al, "Safety of Ketorolac in Neonates and Infants After Cardiac Surgery," *Paediatr Anaesth*, 2006, 16(4):424-8. [PubMed 16618297]

139. Murdoch I, Carver AL, Sultan P, et al. Comparison of different nonsteroidal anti-inflammatory drugs for cesarean section: a systematic review and network meta-analysis. *Korean J Anesthesiol*. 2023;76(6):597-616. doi:10.4097/kja.23014 [PubMed 37066603]

140. National Institute for Health and Care Excellence. Drug allergy: diagnosis and management. Published September 2, 2014. www.nice.org.uk/guidance/cg183

141. Niemi TT, Taxell C, Rosenberg PH. Comparison of the effect of intravenous ketoprofen, ketorolac and diclofenac on platelet function in volunteers. *Acta Anaesthesiol Scand*. 1997;41(10):1353-1358. doi:10.1111/j.1399-6576.1997.tb04657.x [PubMed 9422305]

142. Nissen SE, Yeomans ND, Solomon DH, et al; PRECISION Trial Investigators. Cardiovascular safety of celecoxib, naproxen, or ibuprofen for arthritis. *N Engl J Med*. 2016;375(26):2519-2529. doi:10.1056/NEJMoa1611593 [PubMed 27959716]

143. Oates JA, FitzGerald GA, Branch RA, Jackson EK, Knapp HR, Roberts LJ 2nd. Clinical implications of prostaglandin and thromboxane A2 formation (2). *N Engl J Med*. 1988;319(12):761-767. doi:10.1056/NEJM198809223191206 [PubMed 3045551]

144. Oliveri L, Jerzewski K, Kulik A. Black box warning: is ketorolac safe for use after cardiac surgery? *J Cardiothorac Vasc Anesth*. 2014;28(2):274-279. doi:10.1053/j.jvca.2013.07.014 [PubMed 24231193]

145. O'Mahony D, Cherubini A, Guiteras AR, et al. STOPP/START criteria for potentially inappropriate prescribing in older people: version 3. *Eur Geriatr Med*. 2023;14(4):625-632. doi:10.1007/s41999-023-00777-y [PubMed 37256475]

146. Otani K, Tanigawa T, Watanabe T, et al. Microbiota plays a key role in non-steroidal anti-inflammatory drug-induced small intestinal damage. *Digestion*. 2017;95(1):22-28. doi:10.1159/000452356. [PubMed 28052268]

147. Page RL 2nd, O'Bryant CL, Cheng D, et al; American Heart Association Clinical Pharmacology and Heart Failure and Transplantation Committees of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing; and Council on Quality of Care and Outcomes Research. Drugs that may cause or exacerbate heart failure: A scientific statement from the American Heart Association. *Circulation*. 2016;134(6):e32-e69. doi:10.1161/CIR.0000000000000426. Erratum in: *Circulation*. 2016;134(12):e261. [PubMed 27400984]

148. Papatheodoridis GV, Sougioultzis S, Archimandritis AJ. Effects of *Helicobacter pylori* and nonsteroidal anti-inflammatory drugs on peptic ulcer disease: a systematic review. *Clin Gastroenterol Hepatol*. 2006;4(2):130-142. doi:10.1016/j.cgh.2005.10.006 [PubMed 16469671]

149. Penk JS, Lefaiver CA, Brady CM, Steffensen CM, Wittmayer K. Intermittent versus continuous and intermittent medications for pain and sedation after pediatric cardiothoracic surgery; a randomized controlled trial. *Crit Care Med*. 2018;46(1):123-129. doi:10.1097/CCM.0000000000002771 [PubMed 29028762]

150. Piper JM, Ray WA, Daugherty JR, Griffin MR. Corticosteroid use and peptic ulcer disease: role of nonsteroidal anti-inflammatory drugs. *Ann Intern Med*. 1991;114(9):735-740. doi:10.7326/0003-4819-114-9-735 [PubMed 2012355]

151. Platt E, Neidhardt JM, End B, et al. Safety and efficacy of low-dose versus high-dose parenteral ketorolac for acute pain relief in patients 65 years and older in the emergency department. *Cureus*. 2023;15(6):e40333. doi:10.7759/cureus.40333 [PubMed 37456458]

152. Quan DJ, Kayser SR. Ketorolac induced acute renal failure following a single dose. *J Toxicol Clin Toxicol*. 1994;32(3):305-309. doi:10.3109/15563659409017963 [PubMed 8007038]

153. Rahman S, Malcoun A. Nonsteroidal antiinflammatory drugs, cyclooxygenase-2, and the kidneys. *Prim Care*. 2014;41(4):803-821. doi:10.1016/j.pop.2014.09.001 [PubMed 25439535]

154. Randi ML, Tison T, Luzzatto G, Girolami A. Haemolytic uraemic syndrome during treatment with ketorolac trometamol. *BMJ*. 1993;306(6871):186. doi:10.1136/bmj.306.6871.186-b [PubMed 8443483]

155. Ray WA, Varas-Lorenzo C, Chung CP, et al. Cardiovascular risks of nonsteroidal antiinflammatory drugs in patients after hospitalization for serious coronary heart disease. *Circ Cardiovasc Qual Outcomes*. 2009;2(3):155-163. doi:10.1161/CIRCOUTCOMES.108.805689 [PubMed 20031832]

156. Ready LB, Brown CR, Stahlgren LH, et al. Evaluation of intravenous ketorolac administered by bolus or infusion for treatment of postoperative pain. A double-blind, placebo-controlled, multicenter study. *Anesthesiology*. 1994;80(6):1277-1286. [PubMed 8010474]

157. Reece-Stremtan S, Campos M, Kokajko L; Academy of Breastfeeding Medicine. ABM Clinical Protocol #15: Analgesia and anesthesia for the breastfeeding mother, revised 2017. *Breastfeed Med*. 2017;12(9):500-506. doi:10.1089/bfm.2017.29054.srt [PubMed 29624435]

158. Refer to manufacturer’s labeling.

159. Refer to Canadian labeling.

160. Reinhart DI. Minimising the adverse effects of ketorolac. *Drug Saf*. 2000;22(6):487-497. doi:10.2165/00002018-200022060-00007 [PubMed 10877042]

161. RuDusky BM. Severe postoperative hemorrhage attributed to single-dose parenteral ketorolac-induced coagulopathy. *Angiology*. 2000;51(12):999-1002. doi:10.1177/000331970005101203 [PubMed 11133001]

162. Runyon BA; AASLD. Introduction to the revised American Association for the Study of Liver Diseases practice guideline management of adult patients with ascites due to cirrhosis 2012. *Hepatology*. 2013;57(4):1651-1653. doi:10.1002/hep.26359 [PubMed 23463403]

163. Ruschitzka F, Borer JS, Krum H, et al. Differential blood pressure effects of ibuprofen, naproxen, and celecoxib in patients with arthritis: the PRECISION-ABPM (Prospective Randomized Evaluation of Celecoxib Integrated Safety Versus Ibuprofen or Naproxen Ambulatory Blood Pressure Measurement) Trial. *Eur Heart J*. 2017;38(44):3282-3292. doi:10.1093/eurheartj/ehx508 [PubMed 29020251]

164. Sahu N, Roy J, Vunnam R, Golamari R, Jain R. Naproxen-induced thrombocytopenia. *Proc (Bayl Univ Med Cent)*. 2020;33(4):653-654. doi:10.1080/08998280.2020.1798724 [PubMed 33100559]

165. Sanford-Driscoll M, Knodel LC. Induction of hemolytic anemia by nonsteroidal antiinflammatory drugs. *Drug Intell Clin Pharm*. 1986;20(12):925-934. doi:10.1177/106002808602001202 [PubMed 3545733]

166. Scala E, Giani M, Pirrotta L, et al. Selective severe anaphylactic reaction due to ketorolac tromethamine without nonsteroidal anti-inflammatory drug intolerance. *J Allergy Clin Immunol*. 2001;107(3):557. doi:10.1067/mai.2001.113241 [PubMed 11240963]

167. Schjerning O, Larsen TB, Damkier P. The impact of selective and non-selective non-steroid anti-inflammatory drugs on secondary hemostasis in healthy volunteers. *Thromb Res*. 2009;124(2):208-212. doi:10.1016/j.thromres.2009.01.017 [PubMed 19269020]

168. Schjerning Olsen AM, Fosbøl EL, Lindhardsen J, et al. Duration of treatment with nonsteroidal anti-inflammatory drugs and impact on risk of death and recurrent myocardial infarction in patients with prior myocardial infarction: a nationwide cohort study. *Circulation*. 2011;123(20):2226-2235. doi:10.1161/CIRCULATIONAHA.110.004671 [PubMed 21555710]

169. Schjerning Olsen AM, Gislason GH, McGettigan P, et al. Association of NSAID use with risk of bleeding and cardiovascular events in patients receiving antithrombotic therapy after myocardial infarction. *JAMA*. 2015;313(8):805-814. doi:10.1001/jama.2015.0809 [PubMed 25710657]

170. Schmidt M, Lamberts M, Olsen AM, et al. Cardiovascular safety of non-aspirin non-steroidal anti-inflammatory drugs: review and position paper by the working group for Cardiovascular Pharmacotherapy of the European Society of Cardiology. *Eur Heart J*. 2016;37(13):1015-1023. doi:10.1093/eurheartj/ehv505 [PubMed 26984863]

171. Schneider V, Lévesque LE, Zhang B, Hutchinson T, Brophy JM. Association of selective and conventional nonsteroidal antiinflammatory drugs with acute renal failure: A population-based, nested case-control analysis. *Am J Epidemiol*. 2006;164(9):881-889. doi:10.1093/aje/kwj331 [PubMed 17005625]

172. Schwedt TJ, Garza Ivan. Acute treatment of migraine in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed August 12, 2021.

173. Schwenk ES. Nonopioid pharmacotherapy for acute pain in adults. Post TW, ed. UpToDate. Waltham, MA: UpToDate Inc. http://www.uptodate.com. Accessed October 11, 2022.

174. Schwinghammer AJ, Isaacs AN, Benner RW, Freeman H, O'Sullivan JA, Nisly SA. Continuous infusion ketorolac for postoperative analgesia following unilateral total knee arthroplasty. *Ann Pharmacother*. 2017;51(6):451-456. [PubMed 28478713]

175. Shahin L, Rogers A, Swain L, Ourn FF, Barber T. A case of ketorolac-induced aseptic meningitis. *Cureus*. 2024;16(3):e57349. doi:10.7759/cureus.57349 [PubMed 38690497]

176. Shapiro N. Acute angioedema after ketorolac ingestion: report of case. *J Oral Maxillofac Surg*. 1994;52(6):626-627. doi:10.1016/0278-2391(94)90103-1 [PubMed 8189303]

177. Singer AJ, Mynster CJ, McMahon BJ. The effect of IM ketorolac tromethamine on bleeding time: a prospective, interventional, controlled study. *Am J Emerg Med*. 2003;21(5):441-443. doi:10.1016/s0735-6757(03)00100-1 [PubMed 14523887]

178. Solomon DH, Husni ME, Libby PA, et al. The risk of major NSAID toxicity with celecoxib, ibuprofen, or naproxen: A secondary analysis of the PRECISION trial. *Am J Med*. 2017;130(12):1415-1422.e4. doi:10.1016/j.amjmed.2017.06.028 [PubMed 28756267]

179. Sternlieb P, Robinson RM. Stevens-Johnson syndrome plus toxic hepatitis due to ibuprofen. *N Y State J Med.* 1978;78(8):1239-1243. [PubMed 276660]

180. Strate LL, Gralnek IM. ACG clinical guideline: management of patients with acute lower gastrointestinal bleeding. *Am J Gastroenterol*. 2016;111(4):459-474. doi: 10.1038/ajg.2016.41. [PubMed 26925883]

181. Strom BL, Berlin JA, Kinman JL, et al. Parenteral ketorolac and risk of gastrointestinal and operative site bleeding. A postmarketing surveillance study. *JAMA*. 1996;275(5):376-382. [PubMed 8569017]

182. Strom BL, Carson JL, Schinnar R, Snyder ES, Shaw M, Lundin FE Jr. Nonsteroidal anti-inflammatory drugs and neutropenia. *Arch Intern Med*. 1993;153(18):2119-2124. [PubMed 8379803]

183. Szeto CC, Sugano K, Wang JG, et al. Non-steroidal anti-inflammatory drug (NSAID) therapy in patients with hypertension, cardiovascular, renal or gastrointestinal comorbidities: joint APAGE/APLAR/APSDE/APSH/APSN/PoA recommendations. *Gut*. 2020;69(4):617-629. doi:10.1136/gutjnl-2019-319300 [PubMed 31937550]

184. Taggart E, Doran S, Kokotillo A, Campbell S, Villa-Roel C, Rowe BH. Ketorolac in the treatment of acute migraine: a systematic review. *Headache*. 2013;53(2):277-287. doi:10.1111/head.12009 [PubMed 23298250]

185. Teigen NC, Sahasrabudhe N, Doulaveris G, et al. Enhanced recovery after surgery at cesarean delivery to reduce postoperative length of stay: A randomized controlled trial. *Am J Obstet Gynecol*. 2020;222(4):372.e1–372.e10. [PubMed 31669738]

186. Thorell A, MacCormick AD, Awad S, et al. Guidelines for perioperative care in bariatric surgery: Enhanced Recovery After Surgery (ERAS) Society recommendations. *World J Surg*. 2016;40(9):2065-2083. doi: 10.1007/s00268-016-3492-3. [PubMed 26943657]

187. Toradol tablet (ketorolac) [product monograph]. Vaughan, Ontario, Canada: AA Pharma Inc; August 2024.

188. Toradol IM (ketorolac) [product monograph]. Brantford, Ontario, Canada: Methapharm Inc; April 2023.

189. US Food and Drug Administration (FDA). FDA Drug Safety Communication: FDA strengthens warning that non-aspirin nonsteroidal anti-inflammatory drugs (NSAIDs) can cause heart attacks or strokes. 2015. https://www.fda.gov/drugs/drug-safety-and-availability/fda-drug-safety-communication-fda-strengthens-warning-non-aspirin-nonsteroidal-anti-inflammatory.

190. US Food and Drug Administration (FDA). FDA recommends avoiding use of NSAIDs in pregnancy at 20 weeks or later because they can result in low amniotic fluid. https://www.fda.gov/drugs/drug-safety-and-availability/fda-recommends-avoiding-use-nsaids-pregnancy-20-weeks-or-later-because-they-can-result-low-amniotic. Published October 15, 2020. Accessed October 20, 2020.

191. Välitalo PA, Kemppainen H, Kulo A, et al. Body weight, gender and pregnancy affect enantiomer-specific ketorolac pharmacokinetics. *Br J Clin Pharmacol*. 2017;83(9):1966-1975. [PubMed 28429492]

192. Villanueva M, Heckenberger R, Palmér M, Schrör K. Stereospecific and non-stereospecific effects of ibuprofen on human platelet and polymorphonuclear leukocyte functions. *Agents Actions Suppl*. 1992;37:162-170. doi:10.1007/978-3-0348-7262-1_22 [PubMed 1321554]

193. Walker JJ, Johnstone J, Lloyd J, et al, ”Transfer of Ketorolac Tromethamine From Maternal to Foetal Blood,” *Eur J Clin Pharmacol*, 1988, 34(5):509-11. [PubMed 3264528]

194. Warner TD, Giuliano F, Vojnovic I, Bukasa A, Mitchell JA, Vane JR. Nonsteroid drug selectivities for cyclo-oxygenase-1 rather than cyclo-oxygenase-2 are associated with human gastrointestinal toxicity: a full in vitro analysis. *Proc Natl Acad Sci U S A*. 1999;96(13):7563-7568. doi:10.1073/pnas.96.13.7563. Erratum in: *Proc Natl Acad Sci U S A.* 1999;96(17):9666 [PubMed 10377455]

195. Warth LC, Noiseux NO, Hogue MH, Klaassen AL, Liu SS, Callaghan JJ. Risk of acute kidney injury after primary and revision total hip arthroplasty and total knee arthroplasty using a multimodal approach to perioperative pain control including ketorolac and celecoxib. *J Arthroplasty*. 2016;31(1):253-255. doi:10.1016/j.arth.2015.08.012 [PubMed 26377377]

196. Watcha MF, Ramirez-Ruiz M, White PF, Jones MB, Lagueruela RG, Terkonda RP. Perioperative effects of oral ketorolac and acetaminophen in children undergoing bilateral myringotomy. *Can J Anaesth*. 1992;39(7):649-654. [PubMed 1394752]

197. Weersink RA, Taxis K, Drenth JPH, Houben E, Metselaar HJ, Borgsteede SD. Prevalence of drug prescriptions and potential safety in patients with cirrhosis: a retrospective real-world study. *Drug Saf*. 2019;42(4):539-546. doi:10.1007/s40264-018-0744-1 [PubMed 30357649]

198. Whelton A. Nephrotoxicity of nonsteroidal anti-inflammatory drugs: physiologic foundations and clinical implications. *Am J Med*. 1999;106(5B):13S-24S. doi:10.1016/s0002-9343(99)00113-8 [PubMed 10390124]

199. White AA, Stevenson DD. Aspirin-exacerbated respiratory disease. *N Engl J Med*. 2018;379(11):1060-1070. doi:10.1056/NEJMra1712125 [PubMed 30207919]

200. Wischnik A, Manth SM, Lloyd J, Bullingham R, Thompson JS. The excretion of ketorolac tromethamine into breast milk after multiple oral dosing. *Eur J Clin Pharmacol*. 1989;36(5):521-524. doi:10.1007/BF00558080 [PubMed 2787750]

201. Witting MD. Renal papillary necrosis following emergency department treatment of migraine. *J Emerg Med*. 1996;14(3):373-376. doi:10.1016/0736-4679(96)00036-4 [PubMed 8782036]

202. Yancy CW, Jessup M, Bozkurt B, et al; American College of Cardiology Foundation; American Heart Association Task Force on Practice Guidelines. 2013 ACCF/AHA guideline for the management of heart failure: a report of the American College of Cardiology Foundation/American Heart Association Task Force on Practice Guidelines. *J Am Coll Cardiol*. 2013;62(16):e147-239. doi:10.1016/j.jacc.2013.05.019 [PubMed 23747642]

203. Yeomans ND, Graham DY, Husni ME, et al; PRECISION investigators. Randomised clinical trial: gastrointestinal events in arthritis patients treated with celecoxib, ibuprofen or naproxen in the PRECISION trial. *Aliment Pharmacol Ther*. 2018;47(11):1453-1463. doi:10.1111/apt.14610 [PubMed 29667211]

204. Yousefi H, Sahebi A, Farahani M, Golitaleb M. Anaphylaxis as a rare side effect of ketorolac; a case report. *Arch Acad Emerg Med*. 2020;8(1):e22 [PubMed 32259118]

205. Yurashevich M, Pedro C, Fuller M, Habib AS. Intra-operative ketorolac 15 mg versus 30 mg for analgesia following cesarean delivery: a retrospective study. *Int J Obstet Anesth*. 2020;44:116-121. doi:10.1016/j.ijoa.2020.08.009 [PubMed 32947103]

206. Zipser RD, Hoefs JC, Speckart PF, Zia PK, Horton R. Prostaglandins: modulators of renal function and pressor resistance in chronic liver disease. *J Clin Endocrinol Metab*. 1979;48(6):895-900. doi:10.1210/jcem-48-6-895 [PubMed 447795]

207. Zoubek ME, Lucena MI, Andrade RJ, Stephens C. Systematic review: ibuprofen-induced liver injury. *Aliment Pharmacol Ther*. 2020;51(6):603-611. doi:10.1111/apt.15645 [PubMed 31984540]

Topic 9152 Version 846.0